

From Sweep to Seam: Interleaved Cross-Block Post-Training Quantization

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Abstract

Compressing large language models to two bits or fewer is increasingly feasible through block-wise post-training quantization; cross-block variants reconstruct neighboring Transformer blocks within a moving window. In the fixed two-block setting studied here, the matched sequential baseline moves this window through the network once, so errors introduced early in the sweep are not revisited. We propose Interleaved Cross-Block Quantization (ICBQ), a scheduling modification that revisits the boundary pair between consecutive chunks. Each seam pair is refined twice: first at the end of one chunk and again at the start of the next. The method retains the local two-block objective and reuses the calibration inputs of existing block-wise PTQ pipelines. Under stated local contraction and smoothness assumptions, we derive a depth-wise upper-bound comparison in which seam revisits multiply the propagated term while the residual remains bounded independently of depth. In the reported experiments, ICBQ reduces ternary-quantization perplexity relative to the matched Sequential CBQ baseline, yields finite perplexity in configurations where the baseline has severe degradation, and can also be used with 3-bit and 2-bit GPTQ.

1 Introduction

Deploying large language models (LLMs) at sub-two-bit precision is an important challenge in efficient ML systems. In principle, one could directly optimize the quantized network end to end through quantization-aware training (QAT), as in LLM-QAT and LittleBit [Liu et al., 2024b, Lee et al., 2025]. Alternatively, one could formulate a post-training quantization (PTQ) objective jointly over all blocks. At the scale of modern LLMs, however, such global optimization is rarely feasible. Once models exceed even a few billion parameters, storing the full computation graph, intermediate activations, and optimizer state on a single accelerator typically exceeds the available memory budgets [Ichikawa et al., 2026]. Consequently, practical PTQ methods instead rely on layer- or block-local surrogate objectives [Frantar et al., 2022, Li et al., 2021, Shao et al., 2024, Bulat et al., 2024]. These methods aim to reduce global quantization error by decomposing the problem into a sequence of local subproblems, whose peak memory cost is primarily governed by the reconstruction unit. For large-scale LLMs, block-wise PTQ is a practical route to reducing end-to-end output error within realistic hardware constraints.

Within practical PTQ pipelines, layer-wise weight-only methods such as GPTQ [Frantar et al., 2022] and AWQ [Lin et al., 2024] recover much of the floating-point accuracy at three- to four-bit weight precision, while SmoothQuant [Xiao et al., 2023] targets W8A8 weight-activation PTQ by smoothing activation outliers. Subsequent methods strengthen local objectives in complementary ways: QEP [Arai and Ichikawa, 2025] explicitly propagates and compensates quantization errors; LoqQ [Lin

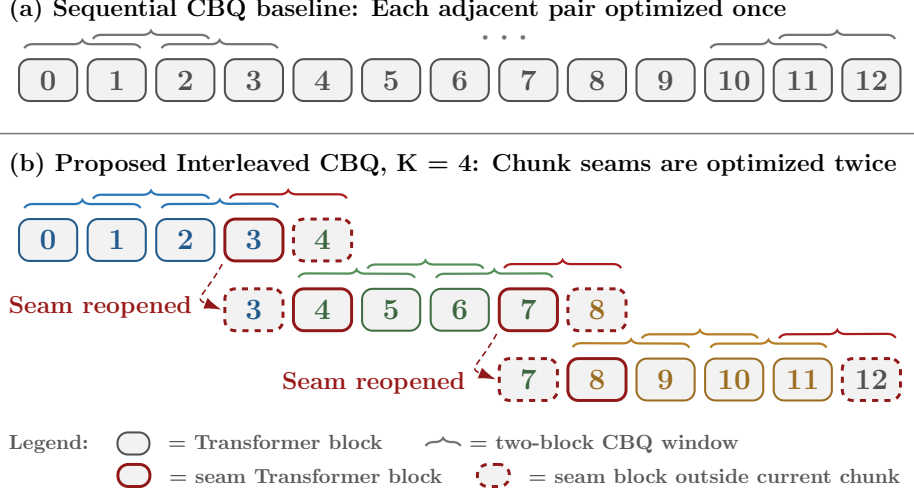


Figure 1: **ICBQ revisits chunk seams using the same calibration samples** ($L = 13$, $K = 4$). **Panel a. Sequential CBQ** performs a single post-hoc sweep, optimizing each adjacent block pair once. **Panel b. Interleaved CBQ** inserts a short CBQ pass after each chunk of newly quantized blocks. Because the final window of chunk c is also the first window of chunk $c+1$, the seam pairs (3, 4), (7, 8), and (11, 12) are optimized twice, whereas non-seam pairs are optimized once. Between chunks, student and teacher activations are recomputed from the stored layer-0 inputs.

and Wan, 2025] introduces output-matching factors within layer-wise PTQ; and LPCD [Ichikawa et al., 2025a] optimizes relaxed objectives over arbitrary submodules before projecting them with layer-wise quantizers. More recently, extreme low-bit methods such as OneBit [Xu et al., 2024], BitNet [Wang et al., 2023], DBF [Boža and Macko, 2026], and MDBF [Ichikawa et al., 2025b] have pushed quantization into increasingly aggressive precision regimes.

CBQ [Ding et al., 2025] introduced cross-block reconstruction to capture dependencies across multiple Transformer blocks. For a controlled schedule comparison, we adapt this principle to a fixed two-block, weight-only reconstruction objective; we refer to the resulting left-to-right baseline as Sequential CBQ for brevity.

Our Sequential CBQ baseline applies each adjacent two-block window once in a left-to-right sweep after all blocks have been quantized. Residual errors introduced at a block pair are then propagated to subsequent blocks without another refinement of that pair. This can increase final activation mismatch with depth. The effect can be pronounced under our ternary DBF setting. For example, Sequential CBQ produces very large perplexities for Llama-3-8B and Qwen3-8B in our setting, motivating a schedule that revisits selected pairs after additional blocks have been processed.

A schedule-level way to mitigate this accumulation is to revisit selected block pairs. We focus on chunk seams: the boundary pairs between consecutive chunks of newly quantized blocks. By interleaving a short CBQ sweep with progressive quantization, the last pair of one chunk also becomes the first pair of the next, so every seam is refined twice, as illustrated in Figure 1. The revisits reuse the same calibration inputs but add refinement computation. We formalize their effect as an additional contraction factor on the propagated term of a depth-wise error bound. Thus, for a network of depth L processed in chunks of size K , the bound reflects the number of chunk seams. We evaluate this schedule with ternary DBF and GPTQ inner quantizers.

2 Preliminaries

Notation and inner quantizers. We consider a Transformer consisting of L blocks. The b -th fp16 teacher block is denoted by f_b^* , where $b \in \{0, \dots, L-1\}$. Given a calibration set $\mathcal{X} = \{x^{(n)}\}_{n=1}^N$, the teacher activations are defined by

$$X_0^{*,(n)} = x^{(n)}, \quad X_{b+1}^{*,(n)} = f_b^*(X_b^{*,(n)}). \quad (1)$$

The corresponding quantized student block is written as $f_b^{\theta_b}$, and the student activations are denoted by $\hat{X}_b^{(n)}$. For each sample, an activation tensor has shape (T, d_{hidden}) , where T is the sequence length and d_{hidden} is the hidden width. Thus, $X_b^{*,(n)}, \hat{X}_b^{(n)} \in \mathbb{R}^{T \times d_{\text{hidden}}}$. Unless otherwise stated, all activation norms are Frobenius norms and are denoted by $\|\cdot\|_F$. We use N exclusively for the number of calibration samples, and reserve $S = S(L, K)$ for the number of chunk seams introduced in Section 3.

An *inner quantizer* is the low-bit routine used to initialize a single Transformer block before any cross-block refinement. We denote it by $f_b^{\theta_b} = Q(f_b^*; \mathcal{Z}_b)$, where $\mathcal{Z}_b = \{Z_b^{(n)}\}_{n=1}^N$ is the set of calibration activations supplied to block b during quantization. The role of Q is local: it receives one fp16 block and produces one low-bit student block whose weights belong to a prescribed codebook. The block-wise and cross-block procedures below do not depend on the specific choice of Q .

Representative choices for Q include Hessian-aware quantizers, such as GPTQ [Frantar et al., 2022], as well as factorized quantizers designed for extreme low-bit settings, such as DBF [Boža and Macko, 2026] and MDBF [Ichikawa et al., 2025b]. The original DBF represents a weight matrix using two binary sign matrices and diagonal scaling matrices. In our experiments, we use a ternary extension in which the factor entries take values in $\{-1, 0, +1\}$; unless otherwise stated, DBF below refers to this ternary implementation. Specifically, for a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, our parameterization is

$$\widehat{W} = D_a A D_m B D_b, \quad D_a = \text{diag}(a), \quad D_m = \text{diag}(m), \quad D_b = \text{diag}(b). \quad (2)$$

where $A \in \{-1, 0, +1\}^{d_{\text{out}} \times k}$ and $B \in \{-1, 0, +1\}^{k \times d_{\text{in}}}$ are ternary factors, $a \in \mathbb{R}_{\geq 0}^{d_{\text{out}}}$, $m \in \mathbb{R}_{\geq 0}^k$, and $b \in \mathbb{R}_{\geq 0}^{d_{\text{in}}}$ are nonnegative scaling vectors, and k denotes the intermediate dimension that determines the compression ratio. We use this ternary DBF implementation as the primary inner quantizer, while GPTQ serves as a complementary baseline for evaluating whether the proposed schedule generalizes beyond a single quantization method.

Block-wise post-training quantization and prefit. Post-training quantization compresses a pre-trained model using only a small calibration set, without retraining on the original training corpus. A natural but impractical objective would be to quantize all Transformer blocks jointly so that the final output of the student model matches that of the fp16 teacher model. For modern LLMs, this global objective is usually too expensive because it requires storing activations and optimization state across many blocks. Block-wise PTQ avoids this cost by decomposing the problem into local subproblems, quantizing one Transformer block at a time [Li et al., 2021, Shao et al., 2024, Frantar et al., 2022].

In a standard block-wise PTQ pipeline, the blocks are processed in depth order. At block b , the inner quantizer Q is applied to the fp16 block f_b^* using the calibration activations available at that depth:

$$f_b^{\theta_b} \leftarrow Q(f_b^*; \mathcal{Z}_b). \quad (3)$$

After block b has been quantized, the student activation stream is advanced by

$$\hat{X}_{b+1}^{(n)} = f_b^{\theta_b}(\hat{X}_b^{(n)}), \quad n = 1, \dots, N. \quad (4)$$

Thus, later blocks are calibrated using activations that already include the effects of earlier quantized blocks. This is the key distinction between block-wise PTQ and purely layer-isolated quantization: the method still optimizes local blocks, but it propagates the student activations through the quantized prefix.

Before applying the inner quantizer, one may optionally perform a *prefit* step. Prefit is a high-precision reconstruction step that adjusts a floating-point copy of the block so that it better matches the fp16 teacher block on the calibration inputs. Given the calibration inputs $\mathcal{Z}_b$ for block b , the prefit solves

$$\phi_b^{\text{pre}} = \arg \min_{\phi_b} \frac{1}{N} \sum_{n=1}^N \left\| f_b^{\phi_b}(Z_b^{(n)}) - f_b^*(Z_b^{(n)}) \right\|_F^2. \quad (5)$$

The resulting high-precision block $f_b^{\phi_b^{\text{pre}}}$ is then passed to the inner quantizer Q . Importantly, prefit itself does not specify a low-bit codebook and is therefore independent of the choice of Q ; it only prepares a better floating-point starting point for the subsequent quantization step. In this work, we enable prefit only when the inner quantizer is ternary DBF.

Cross-block quantization. The limitation of the block-wise procedure above is that each block is judged mainly by how well it reconstructs its own fp16 output. However, in a Transformer, the output error of block i becomes the input error of block $i+1$. A locally small error at the output of one block may therefore be amplified or rotated by the next block, especially at very low precision. CBQ [Ding et al., 2025] addresses this issue through cross-block reconstruction over multiple blocks. To isolate the scheduling effect studied here, we use a simplified weight-only adaptation with a fixed window of two adjacent Transformer blocks, which keeps the memory footprint local.

For the adjacent window $(i, i+1)$, our two-block CBQ-style objective minimizes the reconstruction error at the output of the second block:

$$\min_{\theta_i, \theta_{i+1}} \mathcal{L}_{\text{cbq}}(\theta_i, \theta_{i+1}) = \frac{1}{N} \sum_{n=1}^N \left\| f_{i+1}^{\theta_{i+1}}(f_i^{\theta_i}(\hat{X}_i^{(n)})) - Y_{i+2}^{(n),*} \right\|_F^2, \quad (6)$$

where the fp16 two-block target is

$$Y_{i+2}^{(n),*} = f_{i+1}^*(f_i^*(X_i^{*,(n)})). \quad (7)$$

Here, $\hat{X}_i^{(n)}$ is the current student-side activation entering the window, while $X_i^{*,(n)}$ is the corresponding teacher activation. The objective therefore asks the current student window to reproduce the fp16 teacher output after two consecutive blocks. In practice, we optimize Eq. (6) with a rollback safeguard: if the updated window does not reduce the reconstruction MSE, the parameters are restored to their previous values. The Sequential CBQ baseline used in this paper first initializes all blocks with the inner quantizer Q , and then performs a single left-to-right sweep over the adjacent windows $(0, 1), (1, 2), \dots, (L-2, L-1)$, solving Eq. (6) once for each window and advancing the student activation stream after every step. Thus, throughout this paper, Sequential CBQ denotes this fixed two-block adaptation: Q provides the initial low-bit blocks, and the CBQ-style objective refines neighboring blocks independently of the choice of inner quantizer.

3 Interleaved Cross-Block Quantization

ICBQ keeps the progressive block-wise PTQ pipeline, but partitions the depth axis into consecutive chunks of K blocks and inserts one short interleaved CBQ sweep after each chunk. At each chunk boundary, the refinement includes adjacent pairs around the boundary. For non-final chunks, the right block of the boundary pair can still be unquantized and is handled as a provisional copy until its own quantization step. When the next chunk starts, that same boundary pair is refined again, creating seam overlap that drives seam-amortized contraction and suppresses depth-wise error buildup. After each chunk-level refinement, activation re-roll recomputes student and teacher streams from the saved layer-0 inputs so the next quantization/refinement step sees a coherent state.

3.1 Outer driver

The outer driver is the orchestrator of ICBQ: it decides when each standard block-wise PTQ operation runs and in what order. It sweeps blocks left-to-right ($b = 0, \dots, L-1$), keeps a chunk-start pointer b_{cs} , and for each block applies the usual local quantization step (with optional prefit), then advances the student and teacher streams by one layer through f_b^Q and f_b^* . This follows the standard block-wise PTQ progression from Section 2, but adds explicit chunk-level control over when interleaved cross-block refinement is triggered.

The chunk closes whenever $b+1 - b_{\text{cs}} \geq K$ or $b = L-1$, i.e., once the current chunk reaches size K (or the model ends). At closure, the orchestrator launches a short interleaved CBQ call (Algorithm 2 in Appendix A) on the window range

$$[w_{\min}, w_{\max}) = [\max(0, b_{\text{cs}} - 1), \min(b+1, L-1)), \quad (8)$$

solving (6) on each pair $(i, i+1)$ with $i = w_{\min}, \dots, w_{\max}-1$ (the upper limit avoids the non-existent pair $(L-1, L)$). After this local refinement, the driver re-rolls activations over the processed prefix and advances to the next chunk; the exact inner-loop and re-roll mechanics are detailed in Section 3.3. It then sets $b_{\text{cs}} \leftarrow b+1$ and continues to the next chunk. In short, the outer driver preserves block-by-block PTQ while interleaving per-chunk CBQ: quantize progressively, refine at chunk boundaries, re-roll, repeat.

3.2 The seam: the structural heart of ICBQ

In block-wise PTQ, quality is decided at adjacent-block interfaces. This subsection isolates the key ICBQ overlap effect at those interfaces: chunk seams are revisited once at the end of one chunk and again at the start of the next. Writing $b_{cs} = (c-1)K$ for chunks indexed by $c = 1, 2, \dots$ (with the final chunk truncated at the last valid pair index), the window range (8) becomes

$$\left[\max(0, (c-1)K-1), \min(cK, L-1) \right), \quad (9)$$

that is, the pairs optimized in chunk c are

$$(i, i+1), \quad i = \max(0, (c-1)K-1), \dots, \min(cK, L-1) - 1. \quad (10)$$

For every non-final chunk boundary $cK < L$, the *last* pair processed in chunk c is $(cK-1, cK)$. When chunk $c+1$ starts, its first valid pair index is again $cK-1$, so the *first* pair processed is the same boundary pair $(cK-1, cK)$. Therefore each seam pair is refined twice (end of chunk c , then start of chunk $c+1$), while non-seam interior pairs are refined once (Figure 1). We call $(cK-1, cK)$ the c -th *seam*, and the number of seams in an L -block model with chunk size K is

$$S(L, K) = \lceil L/K \rceil - 1. \quad (11)$$

This “visited twice” seam structure is stated formally as Lemma B.10 in Appendix B. Intuitively, seams are exactly where block-wise errors are handed from one chunk to the next, so revisiting them is the core mechanism that suppresses depth-wise buildup. Theorem 4.1 quantifies this effect through an extra seam-level contraction on the propagated term.

3.3 Activation re-roll for coherent interleaved refinement

The pair updates themselves follow the same regular CBQ subproblem from Section 2; in ICBQ, they are interleaved after each chunk during ongoing block-wise quantization. Given the window range $[w_{\min}, w_{\max})$ from the outer driver, the inner call performs a local block-wise pair sweep. It first recomputes the student and teacher streams to the window start (through blocks $0, \dots, w_{\min}-1$), then iterates over pair indices $i = w_{\min}, \dots, w_{\max}-1$: it recomputes the teacher target Y_{i+2}^* from f_i^*, f_{i+1}^* , updates (θ_i, θ_{i+1}) by minimizing the two-block loss (6), and advances the student/teacher streams one layer via $f_i^{\theta_i}$ and f_i^* . Pseudocode is given in Appendix A.

Because this sweep is interleaved at chunk boundaries, each chunk-level refinement can update blocks whose outputs are still needed by subsequent chunks, so cached activations can become stale. Re-roll resolves this by recomputing coherent student and teacher streams from saved layer-0 inputs: locally to the window start during the within-chunk sweep and through the processed prefix $0, \dots, b$ at chunk end before the next outer step. This mechanism is encoded by Assumption B.4 and used by Theorem 4.1.

Peak-memory footprint. The only state added by ICBQ is a single constant-size buffer of depth-0 activations,

$$\Delta \text{mem}_{\text{ICBQ}} = N \cdot T \cdot d_{\text{hidden}} \cdot \text{bytes}(\text{dtype}), \quad (12)$$

which depends only on the calibration workload—not on the model depth L or the chunk size K . Since the re-roll is executed after the previous window is released, ICBQ never holds two active windows concurrently, and

$$\text{Peak}_{\text{ICBQ}} \leq \text{Peak}_{\text{CBQ}} + \Delta \text{mem}_{\text{ICBQ}}, \quad (13)$$

an L, K -independent additive constant. At $T = 2048$ in fp16 the buffer is ≈ 4.3 GB with $N = 256$ and $d_{\text{hidden}} = 4096$, below 15% of the ternary DBF peak reported in Appendix C.3. Thus, ICBQ has no depth- or chunk-dependent memory growth beyond the added buffer and can be integrated into CBQ-style progressive PTQ pipelines.

4 Theory: Seam-amortized Error Contraction

This section explains the mechanism behind why interleaving with seam revisits can be more stable than plain sequential CBQ. The interpretation is intentionally worst-case: it compares upper bounds,

not guaranteed realized perplexity on every run. Full assumptions and proofs are in Appendix B. The tracked quantity is the boundary mismatch passed from one block pair to the next: $m_i := \|\hat{X}_{i+1} - X_{i+1}^*\|_F$. The reader can interpret this as “how much activation error is handed from pair $(i-1, i)$ to pair $(i, i+1)$.” We denote by m_i^{out} the mismatch after the CBQ call on $(i, i+1)$, with $m_{-1}^{\text{out}} = 0$ from the shared start $\hat{X}_0 = X_0^*$.

Base behavior away from seams. Under Assumptions B.1–B.4 (Appendix B), each regular two-block CBQ step has a damp-and-inject form: it contracts part of incoming mismatch (through $\tau := \gamma\rho < 1$) and adds local approximation noise (ϵ_{sub}). Formally, Lemma B.9 (Appendix B.4) gives

$$m_i^{\text{out}} \leq \tau m_{i-1}^{\text{out}} + (1+\gamma)\epsilon_{\text{sub}}, \quad m_{-1}^{\text{out}} := 0, \quad (14)$$

for every non-seam pair in either schedule.

Why seams are special. At a seam $(cK-1, cK)$, re-rolling resets the local context from stored inputs, and that seam pair is effectively revisited across adjacent chunks. Intuitively, non-seam pairs get one cleanup pass, while seams get an extra cleanup pass on carried mismatch. This yields

$$m_i^{\text{out}} \leq \gamma\tau m_{i-1}^{\text{out}} + (1+\gamma+\gamma^2)\epsilon_{\text{sub}} \quad \text{at a seam} \quad (15)$$

(Lemma B.11, Appendix B.6). Chaining Eq. (14) with one seam recurrence per seam across the depth gives the theorem below: each seam contributes one additional multiplicative contraction factor in the bound.

Theorem 4.1 (Seam-amortized error contraction; informal, formal: Theorem B.5 in Appendix B.2, proved in Appendix B). *Let e_L^{seq} and e_L^{int} denote the depth- L activation mismatches of Sequential CBQ and one-pass ICBQ with chunk size K , on the same calibration set, the same inner quantizer Q , and the same CBQ optimizer budget. Write $\mathcal{B}_L^{\text{seq}}$ for the depth- L upper bound obtained by unrolling Eq. (14) alone (Appendix B.8). Under Assumptions B.1–B.4 and $\tau < 1$, for every $L \geq 2$ and $1 \leq K \leq L$,*

$$\mathbb{E}\|e_L^{\text{int}}\|_F \leq \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma)\epsilon_{\text{sub}}, \quad S(L, K) = \lceil L/K \rceil - 1, \quad (16)$$

where $C(L, K, \rho, \gamma) \leq 1 + (1+\gamma+\gamma^2)\rho/(1-\tau)$ uniformly in L, K , and the per-seam factor γ is tight at the recurrence level (Proposition B.20).

Corollary 4.2 (Geometric decay in the number of seams; informal, formal: Corollary B.8 in Appendix B.2). *Define the amortization gain $G(L, K) := \mathcal{B}_L^{\text{seq}}/\mathcal{B}_L^{\text{int}}$, where $\mathcal{B}_L^{\text{int}}$ is the right-hand side of Eq. (16). If for some $\delta \geq 0$ the residual term satisfies $C\epsilon_{\text{sub}} \leq \delta\gamma^{S(L,K)}\mathcal{B}_L^{\text{seq}}$, then $G(L, K) \geq \gamma^{-S(L,K)}/(1+\delta)$. Thus, when additive noise is not dominant, per-seam shrink factors compound geometrically: the ICBQ-over-sequential bound ratio is exponential in the number of seams.*

Putting Theorem 4.1 and Corollary 4.2 together gives a simple picture: away from seams, mismatch is damped but new local noise is injected; at seams, ICBQ gets an extra damping action, and these actions compound with depth. Thus, chunk size controls how often this correction mechanism is triggered: smaller K means more seams and more opportunities for bound-level contraction using the same calibration data. Equation (16) is still a bound-vs-bound comparison against the sequential upper bound $\mathcal{B}_L^{\text{seq}}$, not necessarily realized sequential error (Remark B.7). For $L = 32$ and $K = 4$ (so $S = 7$), the measured aggregate $\bar{\gamma} \approx 0.67$ on Llama-2-7B (Appendix B.11) gives $\bar{\gamma}^{-7} \approx 16\times$, while a conservative aggregate maximum $\gamma = 0.92$ gives $\approx 1.8\times$; the empirical range 1.08–14.5 $\times$ in Section 5 lies between these diagnostics.

5 Experiments

Our experiments assess whether ICBQ improves perplexity relative to Sequential CBQ and whether the schedule also applies to a second weight-only inner quantizer. We then isolate chunk-size and prefit effects on a shared model subset (Section 5.4, Tables 3–4). **Formatting note.** In all tables of this section, **bold** marks the strongest value in each comparison group.

Table 1: **Ternary DBF results on seven base models.** ICBQ has lower PPL than Sequential CBQ in every reported PPL comparison and higher or equal average zero-shot accuracy in six of seven cases. FP model values are included for reference.

Metric	FP model			No-Ref. (ternary DBF only)			Seq. CBQ			ICBQ (ours)		
	PPL (Wiki-2)	PPL (C4)	Avg. ZS	PPL (Wiki-2)	PPL (C4)	Avg. ZS	PPL (Wiki-2)	PPL (C4)	Avg. ZS	PPL (Wiki-2)	PPL (C4)	Avg. ZS
<i>Standard models</i>												
Mistral-7B	4.69	7.99	0.7128	276.31	143.22	0.3684	36.5	31.1	0.4586	13.8	16.8	0.5404
Llama-2-7B	4.86	7.27	0.6668	26.6	17.71	0.4856	10.7	13.6	0.5330	9.37	12.5	0.5533
Llama-2-13B	4.35	6.73	0.6924	9.92	11.89	0.5960	7.94	10.48	0.6503	7.36	10.09	0.6501
<i>Deep and non-standard-attention models</i>												
Llama-3.2-3B	9.62	15.63	0.6320	66.94	61.44	0.4482	42.6	40.63	0.4706	37.6	37.01	0.4857
Llama-3-8B	7.37	12.24	0.6977	154.43	73.82	0.4715	42.95	48.98	0.5026	34.97	42.32	0.5215
Qwen3-8B	8.58	13.93	0.6946	> 1e4	> 1e4	0.3495	> 1e4	5752.9	0.4032	25.86	29.9	0.5513
Qwen3-14B	7.58	12.48	0.7292	907.71	484.59	0.3529	61.9	44.82	0.6871	19.29	24.67	0.6871

5.1 Setup

Experimental setup. We evaluate seven publicly released base models spanning 3B to 14B parameters. Every experiment is *weight-only*; activations remain in fp16, and no rotation or activation quantizer is applied. We evaluate ternary DBF (1.58-bit) with prefit enabled, and GPTQ at W3 (three bits, per channel) and W2g128 (two bits, group size 128) with prefit disabled. The chunk size is $K = 4$ unless otherwise stated. We report C4-calibrated results in the main text, as C4 calibration was more stable than WikiText-2 calibration for some non-standard-attention architectures in our setting (Appendix C.2). Our direct benchmark is the matched Sequential CBQ baseline defined above, based on the fixed two-block adaptation of CBQ [Ding et al., 2025]; we additionally report No-Refinement (inner quantizer only). During CBQ/ICBQ refinement, ternary DBF runs update the scaling vectors and ternary factors (the factors through STE), while GPTQ runs update scales/zero-points and integer weights (integer weights through STE). Additional setup details are listed in Appendix C.1.

Metrics. We report perplexity (PPL, $\downarrow$) on C4 test splits, using a standard autoregressive language-model evaluation pipeline. We also report average zero-shot accuracy ($\uparrow$) over 7 standard benchmarks. Benchmark details and full per-task zero-shot values are provided in Appendix C.1 and Appendix D.

5.2 ICBQ improves perplexity over Sequential CBQ

Table 1 shows lower PPL for ICBQ than for Sequential CBQ in every reported comparison across standard and deep/non-standard models. Qwen3-8B illustrates the difference: ICBQ yields finite PPL where Sequential CBQ has divergence or very large PPL. The C4 average zero-shot metric follows the same overall pattern; the exception is Llama-2-13B, where Sequential CBQ exceeds ICBQ by 0.0002. Both schedules share the stability condition $\gamma\rho < 1$ (Proposition B.21), while Theorem 4.1 gives ICBQ an additional $\gamma^{S(L,K)}$ factor on the propagated term when the stated assumptions hold.

5.3 Schedule transfer to GPTQ W3 and W2g128

Table 2 shows that ICBQ attains the best or tied-best PPL on five of seven models at both W3 and W2g128, with one tie at each precision. The remaining cases are split between Sequential CBQ and GPTQ-only. This variation is compatible with the bound view: when the propagated term is less dominant, residual and optimization terms can determine the local ordering. Zero-shot results in Appendix D, Table 7, show a similar aggregate pattern, alongside task-specific exceptions.

5.4 The seam is the story: schedule ablations

Returning to ternary DBF, we isolate two schedule-level factors on the same seven-model set used throughout the main experiments: chunk size K and progressive prefit.

Effect of chunk size. Chunk size is the schedule’s primary control knob: it sets how many seam revisits ICBQ performs. Table 3 reports PPL on C4 while varying $K \in \{2, 4, 8, L\}$. Theorem 4.1

Table 2: **The contraction transfers to GPTQ on the seven-model set.** PPL on C4 ($\downarrow$), weight-only GPTQ inner quantizer, fp16 activations, and no prefit ($T_{\text{pre}} = 0$). Within each bit-width block, lower is better.

Model	W3 (per-channel)			W2g128 (group size 128)		
	GPTQ only	Seq. CBQ	ICBQ (ours)	GPTQ only	Seq. CBQ	ICBQ (ours)
Mistral-7B	9.04	8.77	8.86	13.29	9.74	9.76
Llama-2-7B	8.50	8.25	8.20	12.26	9.28	9.22
Llama-2-13B	7.52	7.39	7.39	10.00	8.64	8.76
Llama-3.2-3B	25.58	18.63	18.18	58.16	24.03	22.37
Llama-3-8B	31.42	15.29	14.63	60.65	18.51	17.37
Qwen3-8B	15.65	15.96	15.61	23.08	17.41	17.01
Qwen3-14B	13.54	13.80	13.63	16.21	14.74	14.61

Table 3: **Effect of chunk size K on the seven-model set** (PPL on C4 ($\downarrow$), ternary DBF). $K = L$ is the no-interleaving sequential baseline.

Base model	$K = L$ (Seq.)	$K = 8$	$K = 4$	$K = 2$
Mistral-7B	31.1	17.33	16.8	14.85
Llama-2-7B	13.6	13.07	12.5	12.43
Llama-2-13B	10.48	10.26	10.09	10.03
Llama-3.2-3B	40.63	38.23	37.01	35.62
Llama-3-8B	48.98	45.26	42.32	39.61
Qwen3-8B	5752.9	34.01	29.90	29.80
Qwen3-14B	44.82	29.02	24.67	25.89

predicts stronger contraction as K shrinks because smaller chunks create more seams. Empirically, PPL improves monotonically or near-monotonically as K decreases.

Effect of prefit. Table 4 isolates the prefit step on the same seven-model set (PPL on C4) by setting $T_{\text{pre}} = 0$ under both schedules. Because prefit can improve each block’s starting point, this comparison tests whether the observed differences are confined to the prefit-enabled setting. Interleaving remains better than Sequential CBQ in the reported no-prefit runs, indicating that the schedule effect is not limited to prefit-enabled quantization.

6 Related Work

We organize prior PTQ work by the axis each method *refines*—block unit, objective, weight code, or basis—and position ICBQ as a *schedule*-level refinement under a fixed pair-level objective.

Refining the block unit and objective. BRECQ [Li et al., 2021], OmniQuant [Shao et al., 2024], and QLLM [Liu et al., 2024a] replace layer-wise with block-wise reconstruction. CBQ [Ding et al., 2025] uses cross-block reconstruction to capture dependencies across multiple blocks; our direct baseline instantiates this principle with fixed adjacent two-block windows. A complementary thread sharpens the local *objective* or accounts for cross-layer effects: QEP [Arai and Ichikawa, 2025] explicitly propagates quantization errors to compensate for their accumulation; LoaQ [Lin and Wan, 2025] introduces output-matching factors to improve output-level approximation within layer-wise PTQ; and LPCD [Ichikawa et al., 2025a] optimizes relaxed objectives over arbitrary submodules and projects the solutions with layer-wise quantizers. ICBQ changes the *schedule* while retaining a fixed pair-level objective; it may be combined with compatible inner objectives.

Refining the weight code and basis. BitNet [Wang et al., 2023], the 1.58-bit line of Ma et al. [2024], DBF [Boža and Macko, 2026], MDBF [Ichikawa et al., 2025b], and OneBit [Xu et al., 2024] sharpen binary or ternary weight representations; ICBQ is agnostic to this axis and uses the selected method as its inner quantizer. QuIP [Chee et al., 2023] applies incoherence processing for low-bit parameter quantization. Rotation-based methods such as QuaRot [Ashkboos et al., 2024] and

Table 4: **Interleaving helps even without prefit.** PPL on C4 ($\downarrow$) with ternary DBF, with and without the prefit step ($T_{\text{pre}} = 50$ vs. 0), on the same seven-model set.

Base model	Prefit + Seq. CBQ	Prefit + Interleaved CBQ (ours)	No-prefit + Seq. CBQ	No-prefit + Interleaved CBQ
Mistral-7B	31.1	16.8	19.71	18.9
Llama-2-7B	13.6	12.5	15.50	15.4
Llama-2-13B	10.48	10.09	11.85	11.80
Llama-3.2-3B	40.63	37.01	51.66	50.00
Llama-3-8B	48.98	42.32	68.53	66.35
Qwen3-8B	5752.9	29.90	47.28	41.59
Qwen3-14B	44.82	24.67	35.30	31.42

SpinQuant [Liu et al., 2025] facilitate low-bit weight–activation inference by reducing outlier effects; these transformations target the quantization basis and may be combined with ICBQ as preprocessing.

Schedule theory beyond quantization. The *order* in which subproblems are solved controls propagated error—a theme in asynchronous block-coordinate descent, multigrid V/W-cycles, and parallel quasi-quantum annealing [Ichikawa and Arai, 2025]. ICBQ applies this perspective to block-wise PTQ by using the chunk seam as a point for an additional refinement with the same calibration samples.

7 Conclusion

Block-wise PTQ has mainly focused on what happens *inside* the reconstruction window; ICBQ also considers *when* the window is applied. Interleaving pair-refinement passes with progressive quantization makes each chunk seam a second refinement point and adds a factor of γ to the propagated term of the depth-wise error upper bound under the stated assumptions (Theorem 4.1). The schedule reuses the calibration samples and preserves the inner objective. In the reported ternary DBF experiments, ICBQ has lower PPL than Sequential CBQ on the seven main models; it also yields finite PPL for Qwen3-8B and Llama-3-8B configurations with severe Sequential-CBQ degradation. The GPTQ results show that the schedule can also be used with W3 and W2g128. These results suggest that schedule design is a useful additional consideration for low-bit weight-only PTQ.

Limitations. Even with interleaving, the method remains block-local, so it does not fully capture the behavior of the truly global end-to-end output error. Bridging this gap—both algorithmically and theoretically—is a key direction for future work. In addition, our experiments focus on language models. Whether the same schedule-level gains transfer to other families (e.g., vision models) remains open.

Broader impacts. Potential positive impact: ICBQ lowers the memory and energy footprint of deploying large language models at sub-two-bit precision, which enables wider access to LLM inference on commodity hardware and reduces the carbon cost of inference-heavy workloads. Potential negative impact: a cheaper deployment pipeline could lower the cost barrier for malicious or low-quality deployment of LLMs; however, ICBQ is a pure quantization schedule that does not alter model behavior beyond reconstruction error, so the risk profile matches that of existing PTQ methods such as GPTQ [Frantar et al., 2022] and CBQ [Ding et al., 2025]. We view the net impact as positive within the standard PTQ risk envelope.

References

- Yamato Arai and Yuma Ichikawa. Quantization error propagation: Revisiting layer-wise post-training quantization. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL <https://arxiv.org/abs/2504.09629>.
- Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Pashmina Cameron, Martin Jaggi, Dan Alistarh, Torsten Hoefer, and James Hensman. QuaRot: Outlier-free 4-Bit inference in rotated LLMs. In *Advances in Neural Information Processing Systems*, volume 37,

2024. doi: 10.52202/079017-3180. URL https://papers.nips.cc/paper_files/paper/2024/hash/b5b939436789f76f08b9d0da5e81af7c-Abstract-Conference.html.
- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. PIQA: Reasoning about physical commonsense in natural language. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05):7432–7439, 2020. doi: 10.1609/aaai.v34i05.6239. URL <https://ojs.aaai.org/index.php/AAAI/article/view/6239>.
- Vladimír Boža and Vladimír Macko. Addition is almost all you need: Compressing large language models with double binary factorization. *Transactions on Machine Learning Research*, 2026. ISSN 2835-8856. URL <https://openreview.net/forum?id=k5kUKoewdQ>.
- Adrian Bulat, Yassine Ouali, and Georgios Tzimiropoulos. QBB: Quantization with binary bases for LLMs. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0105. URL https://papers.nips.cc/paper_files/paper/2024/hash/05b69cc4c8ff6e24c5de1ecd27223d37-Abstract-Conference.html.
- Jerry Chee, Yaohui Cai, Volodymyr Kuleshov, and Christopher M. De Sa. QuIP: 2-bit quantization of large language models with guarantees. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-0196. URL https://papers.nips.cc/paper_files/paper/2023/hash/0df38cd13520747e1e64e5b123a78ef8-Abstract-Conference.html.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 2924–2936. Association for Computational Linguistics, 2019. doi: 10.18653/v1/N19-1300. URL <https://aclanthology.org/N19-1300/>.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018. URL <https://arxiv.org/abs/1803.05457>.
- Xin Ding, Xiaoyu Liu, Zhijun Tu, Yun Zhang, Wei Li, Jie Hu, Hanting Chen, Yehui Tang, Zhiwei Xiong, Baoqun Yin, and Yunhe Wang. CBQ: Cross-block quantization for large language models. In *International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=eW4yh6HKz4>.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022. doi: 10.48550/arXiv.2210.17323. URL <https://arxiv.org/abs/2210.17323>.
- Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, Jason Phang, Laria Reynolds, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation. Zenodo, 2021. URL <https://zenodo.org/records/5371629>.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025. URL <https://arxiv.org/abs/2503.19786>.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Yuma Ichikawa and Yamato Arai. Optimization by parallel quasi-quantum annealing with gradient-based sampling. In *International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=9EfBeXaXf0>.
- Yuma Ichikawa, Yudai Fujimoto, and Akira Sakai. LPCD: Unified framework from layer-wise to submodule quantization. *arXiv preprint arXiv:2512.01546*, 2025a. URL <https://arxiv.org/abs/2512.01546>.
- Yuma Ichikawa, Yoshihiko Fujisawa, Yudai Fujimoto, Akira Sakai, and Katsuki Fujisawa. More than bits: Multi-envelope double binary factorization for extreme quantization. *arXiv preprint arXiv:2512.24545*, 2025b. URL <https://arxiv.org/abs/2512.24545>.

- Yuma Ichikawa, Keiji Kimura, Akihiro Yoshida, Yudai Fujimoto, Hiroki Tokura, Yamato Arai, Yoshiyuki Ishii, Yusei Kawakami, Genki Shikada, Achille Jacquemond, Yoshihiko Fujisawa, Katsuki Fujisawa, Takumi Honda, and Akira Sakai. OneComp: One-line revolution for generative AI model compression. *arXiv preprint arXiv:2603.28845*, 2026. URL <https://arxiv.org/abs/2603.28845>.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023. URL <https://arxiv.org/abs/2310.06825>.
- Banseok Lee, Dongkyu Kim, Youngcheon You, and Youngmin Kim. LittleBit: Ultra low-bit quantization via latent factorization. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://papers.nips.cc/paper_files/paper/2025/hash/a917c1d57088897beba47f96b4fd7f3c-Abstract-Conference.html.
- Yuhang Li, Ruihao Gong, Xu Tan, Yang Yang, Peng Hu, Qi Zhang, Fengwei Yu, Wei Wang, and Shi Gu. BRECQ: Pushing the limit of post-training quantization by block reconstruction. In *International Conference on Learning Representations*, 2021.
- Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *Proceedings of Machine Learning and Systems*, volume 6, 2024. URL https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html.
- Li Lin and Xiaojun Wan. LoqQ: Layer-wise output approximation quantization. *arXiv preprint arXiv:2509.06297*, 2025. URL <https://arxiv.org/abs/2509.06297>.
- Jing Liu, Ruihao Gong, Xiuying Wei, Zhiwei Dong, Jianfei Cai, and Bohan Zhuang. QLLM: Accurate and efficient low-bitwidth quantization for large language models. In *International Conference on Learning Representations*, 2024a. URL <https://openreview.net/forum?id=FIp1mUWdm3>.
- Zechun Liu, Barlas Oguz, Changsheng Zhao, Ernie Chang, Pierre Stock, Yashar Mehdad, Yangyang Shi, Raghuraman Krishnamoorthi, and Vikas Chandra. LLM-QAT: Data-free quantization aware training for large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 467–484. Association for Computational Linguistics, 2024b. doi: 10.18653/v1/2024.findings-acl.26. URL <https://aclanthology.org/2024.findings-acl.26/>.
- Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. SpinQuant: LLM quantization with learned rotations. In *International Conference on Learning Representations*, 2025.
- Shuming Ma, Hongyu Wang, Lingxiao Ma, Lei Wang, Wenhui Wang, Shaohan Huang, Li Dong, Ruiping Wang, Jilong Xue, and Furu Wei. The era of 1-Bit LLMs: All large language models are in 1.58 bits. *arXiv preprint arXiv:2402.17764*, 2024. URL <https://arxiv.org/abs/2402.17764>.
- Meta Llama Team. Llama 3.2 model card. https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD.md, 2024. Accessed: 2026-05-01.
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. Can a suit of armor conduct electricity? a new dataset for open book question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2381–2391. Association for Computational Linguistics, 2018. doi: 10.18653/v1/D18-1260. URL <https://aclanthology.org/D18-1260/>.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. WinoGrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9):99–106, 2021. doi: 10.1145/3474381.
- Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Zhao, Zhiqian Li, Kaipeng Zhang, Peng Gao, Yu Qiao, and Ping Luo. OmniQuant: Omnidirectionally calibrated quantization for large language models. In *International Conference on Learning Representations*, 2024.
- Hugo Touvron, Louis Martin, Kevin Stone, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023. URL <https://arxiv.org/abs/2307.09288>.

- Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Haoran Wang, Lingxiao Ma, Fan Yang, Ruiping Wang, Yi Wu, and Furu Wei. BitNet: Scaling 1-Bit transformers for large language models. *arXiv preprint arXiv:2310.11453*, 2023. URL <https://arxiv.org/abs/2310.11453>.
- Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 38087–38099. PMLR, 2023. URL <https://proceedings.mlr.press/v202/xiao23c.html>.
- Yuzhuang Xu, Xu Han, Zonghan Yang, Shuo Wang, Qingfu Zhu, Zhiyuan Liu, Weidong Liu, and Wanxiang Che. OneBit: Towards extremely low-bit large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2122. URL https://papers.nips.cc/paper_files/paper/2024/hash/7a7a3f53faafc0161be0fcb57e5fa078-Abstract-Conference.html.
- An Yang, Anfeng Li, Baosong Yang, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4791–4800. Association for Computational Linguistics, 2019. doi: 10.18653/v1/P19-1472. URL <https://aclanthology.org/P19-1472/>.

Appendix Contents

- Appendix A: algorithm and schedule details.
- Appendix B: formal assumptions, theorem statements, proofs, and related theory diagnostics.
- Appendix C: experimental and implementation details.
- Appendix D: supplementary empirical results.

Notation

Symbol	Meaning
L	Number of transformer blocks, indexed by $b \in \{0, \dots, L-1\}$.
K	Chunk size used by ICBQ.
$S(L, K)$	Number of chunk seams, $S(L, K) = \lceil L/K \rceil - 1$.
$cK-1$	Pair index of the c -th seam, corresponding to boundary pair $(cK-1, cK)$.
f_b^*	The fp16 teacher block at depth b .
$f_b^{\theta_b}$	The current student block at depth b with parameter state θ_b .
Q	Inner one-block weight-only quantizer, instantiated as DBF or GPTQ.
X_b^*	Teacher activation at depth b on the calibration sample under discussion.
$\hat{X}_b$	Student activation at depth b .
e_b	Depth- b activation mismatch, $e_b = \hat{X}_b - X_b^*$.
m_i	Midpoint residual of pair $(i, i+1)$, $m_i = \ \hat{X}_{i+1} - X_{i+1}^*\ _F$.
u_i, v_i	Post-pair midpoint residuals for Sequential CBQ and ICBQ, respectively.
$\mathcal{T}_b$	Finite set of teacher and student activations at depth b visited by either schedule.
Θ_b	Finite set of parameter states assumed by block b during either schedule.
ρ	Uniform teacher-block Lipschitz constant on the visited activation sets.
γ	Per-sample midpoint contraction factor certified for each actual CBQ call.
ϵ_{sub}	Common residual envelope covering local representation error and CBQ fixed-point error.
τ	Shorthand $\tau = \gamma\rho$; the proof assumes $\tau < 1$.
$R, \tilde{R}$	Recurrence forcing constants, $R = (1+\gamma)\epsilon_{\text{sub}}$ and $\tilde{R} = (1+\gamma+\gamma^2)\epsilon_{\text{sub}}$.
$s(i)$	Number of seam indices at most i .
$\mathcal{U}_i, \mathcal{V}_i$	Deterministic upper bounds on sequential and interleaved post-pair midpoint residuals.
$\mathcal{B}_L^{\text{seq}}, \mathcal{B}_L^{\text{int}}$	Sequential and interleaved depth- L error upper bounds.
$C(L, K, \rho, \gamma)$	Additive residual constant in the formal ICBQ depth- L bound.

A Algorithm and Schedule Details

We record the full ICBQ schedule as two algorithms. The outer driver (Algorithm 1) implements the progressive weight-only quantization loop: for each block it optionally runs `prefit`, applies the inner quantizer Q , advances the activation streams, and at every chunk boundary calls the inner refinement followed by an end-of-chunk re-roll. The inner refinement (Algorithm 2) implements the interleaved CBQ call: given the window range $[w_{\min}, w_{\max})$ produced by the outer driver it starts with an in-pass re-roll through blocks $0, \dots, w_{\min}-1$ and then sweeps through the valid pair indices, running at each pair a rollback-safeguarded Adam update on the plain-MSE two-block CBQ loss $\mathcal{L}_{\text{cbq}}$ followed by a one-layer advance of the student and teacher streams. Here $\mathcal{L}_{\text{pre}}$ denotes the one-block DBF `prefit` loss, and $\mathcal{L}_{\text{cbq}}$ denotes the plain-MSE two-block reconstruction loss used in our CBQ-style adaptation.

Algorithm 1 Interleaved Cross-Block Quantization (outer progressive driver).

Require: calibration batches $\mathcal{X}$, teacher blocks $\{f_b^*\}_{b=0}^{L-1}$, inner quantizer Q , chunk size K , prefit steps T_{pre} , inner-quantizer budget T_Q .

- 1: **Save** stored inputs $\{X_0^{(s)}\}_s$.
 - 2: Initialize $\hat{X}_0 \leftarrow X_0^* \leftarrow \{X_0^{(s)}\}_s$, $b_{\text{cs}} \leftarrow 0$.
 - 3: **for** $b = 0, \dots, L - 1$ **do**
 - 4: **if** $Q = \text{DBF}$ **then**
 - 5: **Prefit:** run T_{pre} AdamW steps on $\mathcal{L}_{\text{pre}}(\theta_b)$ in float32.
 - 6: **end if**
 - 7: **Quantize:** run the inner quantizer Q with budget T_Q on block b .
 - 8: Advance: $\hat{X}_{b+1} \leftarrow f_b^Q(\hat{X}_b)$; $X_{b+1}^* \leftarrow f_b^*(X_b^*)$.
 - 9: **if** $b + 1 - b_{\text{cs}} \geq K$ **or** $b = L - 1$ **then**
 - 10: $w_{\min} \leftarrow \max(0, b_{\text{cs}} - 1)$, $w_{\max} \leftarrow \min(b + 1, L - 1)$
 - 11: Call Algorithm 2 with $[w_{\min}, w_{\max}]$.
 - 12: **Re-roll:** recompute $\hat{X}_{0 \rightarrow b+1}$ and $X_{0 \rightarrow b+1}^*$ from stored inputs.
 - 13: $b_{\text{cs}} \leftarrow b + 1$.
 - 14: **end if**
 - 15: **end for**
 - 16: **return** quantized model $\{f_b^Q\}_b$ with refined parameters.
-

Algorithm 2 Interleaved CBQ refinement on a window range $[w_{\min}, w_{\max}]$.

Require: Student state $\{f_b^{\theta_b}\}_{b \leq w_{\max}}$, teacher state $\{f_b^*\}_{b \leq w_{\max}}$, stored calibration inputs $\{X_0^{(s)}\}_s$.

- 1: Re-roll: advance student and teacher copies in lockstep through blocks $0, \dots, w_{\min} - 1$ from the stored inputs.
 - 2: **for** $i = w_{\min}, \dots, w_{\max} - 1$ **do**
 - 3: Compute teacher targets $\{Y_{i+2}^{(s),*}\}_s$ by passing the current teacher state through f_i^*, f_{i+1}^* .
 - 4: Minimize the two-block CBQ loss $\mathcal{L}_{\text{cbq}}$ on (θ_i, θ_{i+1}) with Adam; rollback on non-improvement.
 - 5: Advance student and teacher by one layer (through $f_i^{\theta_i}$ and f_i^*).
 - 6: **end for**
-

B Formal Theory and Proofs

This appendix collects the three structural hypotheses (Assumptions B.1, B.2, and B.4), two auxiliary remarks, and the formal theorem and corollary statements used by the detailed proof below. These formal statements correspond to the informal seam-amortized contraction theorem and amortization-gain corollary in the main paper.

B.1 Assumptions

Preamble. Fix a calibration set $\mathcal{X} = \{x^{(n)}\}_{n=1}^N$ and a chunk size K . Let $\mathcal{S}_{\text{seq}}$ denote the fixed two-block Sequential CBQ adaptation defined in Section 2, and let $\mathcal{S}_{\text{int}}$ denote ICBQ with this chunk size; both schedules are applied to the same calibration set, use the same fp16 teacher blocks $\{f_b^*\}_{b=0}^{L-1}$, and use the same inner quantizer Q (including the same DBF prefit objective $\mathcal{L}_{\text{pre}}$ when $Q = \text{DBF}$) and the same rollback-safeguarded Adam budget for every CBQ subproblem. The constants $\rho, \gamma, \epsilon_{\text{sub}}$ below are certified for the schedules at this fixed K ; the theorem therefore applies pointwise to any K for which these certificates hold. A claim uniform over multiple chunk sizes requires the same constants to hold uniformly over those choices of K . Both schedules execute deterministic, finitely many forward passes and parameter updates; hence, for each depth b , the set of depth- b student activations $\hat{X}_b^{(n)}$ arising anywhere in either schedule is finite. Denote

$$\mathcal{T}_b := \{X_b^{*,(n)} : n \in [N]\} \cup \{\hat{X}_b^{(n)} \text{ arising in } \mathcal{S}_{\text{seq}}\} \cup \{\hat{X}_b^{(n)} \text{ arising in } \mathcal{S}_{\text{int}}\} \subset \mathbb{R}^{T \times d_{\text{hidden}}}. \quad (17)$$

Assumption B.1 (Lipschitz teacher blocks). *There exists a constant $\rho > 0$, independent of the depth b and of the schedule, such that*

$$\|f_b^*(x) - f_b^*(y)\|_F \leq \rho \|x - y\|_F \text{ for all } x, y \in \mathcal{T}_b, b \in \{0, \dots, L-1\}. \quad (18)$$

The non-expansive regime $\rho \leq 1$ captures empirical residual-stream behavior of trained transformer blocks; our proofs only require $\tau := \gamma\rho < 1$, so locally expansive blocks ($\rho > 1$) are admissible if γ is small (Appendix B.10).

For each depth i , let Θ_i denote the finite set of parameter states assumed by block i at any intermediate state of either schedule (initial fp16 teacher parameters, post-prefit, post-inner-quantizer, post-CBQ-update, and any other pre-pass state that appears as an input to a CBQ subproblem).

Assumption B.2 (Contractive two-block CBQ subproblem). *There exist constants $\gamma \in (0, 1)$ and $\epsilon_{\text{sub}} \geq 0$, independent of the depth and of the schedule, such that:*

(Local representation envelope.) *For every i , every $\theta_i \in \Theta_i$, and every $z \in \mathcal{T}_i$,*

$$\|f_i^{\theta_i}(z) - f_i^*(z)\|_F \leq \epsilon_{\text{sub}}. \quad (19)$$

(Midpoint contraction.) *For every actual CBQ call made by either schedule on a pair $(i, i+1)$, let $(\theta_i^{\text{pre}}, \theta_{i+1}^{\text{pre}})$ be the pre-pass block states and $\{\hat{X}_i^{(n)}\}_{n=1}^N$ the student inputs; let $(\theta_i^{\text{post}}, \theta_{i+1}^{\text{post}})$ be the shared post-pass pair returned by the rollback-safeguarded Adam optimizer applied to the plain-MSE two-block CBQ loss $\mathcal{L}_{\text{cbq}}$. Then, for every sample n and every $\hat{X}_i = \hat{X}_i^{(n)}$, writing*

$$r^{\text{in}} := \|f_i^{\theta_i^{\text{pre}}}(\hat{X}_i) - X_{i+1}^*\|_F, \quad r^{\text{out}} := \|f_i^{\theta_i^{\text{post}}}(\hat{X}_i) - X_{i+1}^*\|_F, \quad (20)$$

we have

$$r^{\text{out}} \leq \gamma r^{\text{in}} + \epsilon_{\text{sub}}. \quad (21)$$

Eq. (21) is the linear-convergence form of a first-order method with a nonzero fixed-point residual; Eq. (19) is the local representation envelope needed to transition from a depth- i input error to a depth- $(i+1)$ midpoint error. Three points deserve emphasis. First, Eq. (21) is *per-sample*: the optimizer returns a single parameter pair, but the assumption requires the induced midpoint residual to contract uniformly over every presented sample, which is strictly stronger than contraction on the sample-averaged residual. Second, the rollback safeguard only certifies the non-increase of the monitored two-block loss; the per-sample midpoint contraction in Eq. (21) is an additional empirical certificate (Remark B.3 below). Third, if separate constants are preferred, ϵ_{sub} can be split into a local representation radius ϵ_{rep} (in Eq. (19)) and a CBQ fixed-point radius ϵ_{opt} (in Eq. (21)), and every subsequent inequality holds with ϵ_{sub} replaced by $\max(\epsilon_{\text{rep}}, \epsilon_{\text{opt}})$. The aggregate constant γ is estimated empirically in Appendix B.11.

Remark B.3 (Midpoint contraction is an additional certificate, not a consequence of rollback). *The rollback safeguard is applied to the two-block output reconstruction loss $\mathcal{L}_{\text{cbq}}$, which couples blocks i and $i+1$ jointly through $f_{i+1}^{\theta_{i+1}} \circ f_i^{\theta_i}$. The midpoint residual depends only on block i , so rollback does not by itself imply the per-sample midpoint contraction in Eq. (21). Assumption B.2 is therefore an additional local certificate of the implemented rollback-safeguarded Adam on the finite collection of CBQ calls actually made by the schedules; the proof of Theorem B.5 uses only this midpoint certificate together with the local representation envelope, not the fact that the two-block loss decreases. The empirical study in Appendix B.11 provides an aggregate diagnostic of this behavior, rather than a uniform certificate.*

Assumption B.4 (Exact re-roll). *Every re-roll performed by either schedule has an associated prefix length d : for the in-pass re-roll at the start of every interleaved refinement call (line 1 of Algorithm 2), $d = w_{\text{min}}$; for the end-of-chunk re-roll in the outer driver, performed after block b is processed, $d = b + 1$. The re-roll recomputes, for each calibration sample n and every target depth $b' \in \{0, \dots, d\}$,*

$$\hat{X}_{b'}^{(n)} = (f_{b'-1}^{\theta_{b'-1}} \circ \dots \circ f_0^{\theta_0})(X_0^{(n)}), \quad X_{b'}^{*(n)} = (f_{b'-1}^* \circ \dots \circ f_0^*)(X_0^{(n)}), \quad (22)$$

where $\theta_0, \dots, \theta_{b'-1}$ are the current parameter states of blocks $0, \dots, b'-1$ at the moment of the re-roll (with $b' = 0$ denoting the empty composition). In the same symbolic sense, every single-block forward pass executed elsewhere in either schedule is treated as an exact function application of its block map.

In our implementation, both re-rolls and every in-loop advance step are deterministic forward passes through their current prefixes using the stored layer-0 inputs. Assumption B.4 abstracts away floating-point non-associativity in the symbolic identities used by the proof. The in-pass re-roll identifies the repeated seam input, while the end-of-chunk re-roll keeps the states and activations included in Θ_i and $\mathcal{T}_i$ consistent with the current prefix.

B.2 Formal theorem, corollary, and remarks

For a schedule $\mathcal{S} \in \{\mathcal{S}_{\text{seq}}, \mathcal{S}_{\text{int}}\}$, write $e_b^{\mathcal{S}} := \hat{X}_b^{\mathcal{S}} - X_b^*$ for the depth- b activation mismatch on a calibration sample. The expectation $\mathbb{E}$ in the theorem statements is the empirical average over the fixed calibration set $\mathcal{X}$.

Under Assumptions B.1, B.2, and B.4 and $\tau < 1$, define the *sequential depth- L bound*

$$\mathcal{B}_L^{\text{seq}} := \rho(1 + \gamma) \frac{1 - \tau^{L-1}}{1 - \tau} \epsilon_{\text{sub}} + \epsilon_{\text{sub}}, \quad (23)$$

which is the sequential upper bound used for the comparison (derivation in Appendix B.8).

Theorem B.5 (Seam-amortized error contraction, formal). *Under Assumptions B.1, B.2, and B.4 and $\tau = \gamma\rho < 1$, for every $L \geq 2$ and every $1 \leq K \leq L$,*

$$\mathbb{E} \|e_L^{\text{int}}\|_F \leq \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma) \epsilon_{\text{sub}}, \quad (24)$$

where $S(L, K) = \lceil L/K \rceil - 1$, and one admissible (generally non-tight) choice of the residual constant is

$$\begin{aligned} C(L, K, \rho, \gamma) &= (1 - \gamma^{S(L,K)}) \\ &\quad + \left[(1 + \gamma + \gamma^2) - \gamma^{S(L,K)}(1 + \gamma) \right] \rho \frac{1 - \tau^{L-1}}{1 - \tau}. \end{aligned} \quad (25)$$

In particular,

$$C(L, K, \rho, \gamma) \leq 1 + \frac{(1 + \gamma + \gamma^2)\rho}{1 - \tau} \text{ uniformly in } L, K, \quad (26)$$

and for the standard stability regime $\rho \leq 1$ this simplifies to $C \leq 1 + (1 + \gamma + \gamma^2)/(1 - \tau)$; for $K = L$ the leading factor is $\gamma^0 = 1$ (one chunk, no seams), and for $K = 2$ it contracts by $\gamma^{\lceil L/2 \rceil - 1}$. The per-seam factor γ is tight at the recurrence level (Proposition B.20).

Remark B.6 (Non-tightness of the displayed residual constant). *The constant in Eq. (25) is chosen for a simple closed-form comparison with the sequential bound $\mathcal{B}_L^{\text{seq}}$. It is one admissible upper-bound constant produced by the bookkeeping in Proposition B.14—specifically by the pointwise estimate $\beta_k \leq \tilde{R}$ used in Eq. (49)—and is generally not the smallest constant one could write down. In particular, when $K = L$ there is a single chunk and no seam ($S = 0$), so ICBQ and sequential CBQ coincide as schedules at the recurrence level; the displayed residual constant nevertheless reduces only to $C(L, L, \rho, \gamma) = \gamma^2 \rho (1 - \tau^{L-1}) / (1 - \tau)$ rather than to zero, because the same $\beta_k \leq \tilde{R}$ estimate is applied to non-seam pairs in Eq. (49). A sharper seam-aware comparison that vanishes in the no-seam case is recorded in Proposition B.15 below.*

Remark B.7 (Upper-bound semantics). *The formal ICBQ depth- L bound in Theorem B.5 compares ICBQ’s expected depth- L error against the explicit sequential upper bound $\mathcal{B}_L^{\text{seq}}$ derived from the same recurrences, rather than against the realized sequential error. The comparison is therefore a bound-level statement: the leading propagated term carries the factor γ^S , while all additive terms that do not carry this factor are collected in $C \epsilon_{\text{sub}}$.*

Corollary B.8 (Geometric decay in the number of seams, formal). *When $\mathcal{B}_L^{\text{seq}} > 0$, define the amortization gain of ICBQ over sequential CBQ as $G(L, K) := \mathcal{B}_L^{\text{seq}} / \mathcal{B}_L^{\text{int}}$, where $\mathcal{B}_L^{\text{int}} = \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma) \epsilon_{\text{sub}}$ is the explicit right-hand side of Eq. (24). Under the hypotheses of Theorem B.5, if for some $\delta \geq 0$,*

$$C(L, K, \rho, \gamma) \epsilon_{\text{sub}} \leq \delta \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}}, \quad (27)$$

then

$$G(L, K) \geq \frac{\gamma^{-S(L,K)}}{1 + \delta}. \quad (28)$$

Consequently, for any family of instances in which $C(L, K, \rho, \gamma) \epsilon_{\text{sub}} = o(\gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}})$,

$$G(L, K) \geq \gamma^{-S(L,K)} (1 - o(1)) = \gamma^{-(\lceil L/K \rceil - 1)} (1 - o(1)). \quad (29)$$

Proof. By Theorem B.5, the interleaved upper bound is $\mathcal{B}_L^{\text{int}} = \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma) \epsilon_{\text{sub}}$ with $C \geq 0$, so $\mathcal{B}_L^{\text{int}} \leq (1 + \delta) \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}}$ under the hypothesis, and $\mathcal{B}_L^{\text{int}} \geq \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} > 0$. Dividing yields $G(L, K) \geq 1/((1 + \delta) \gamma^{S(L,K)})$, which is Eq. (28). The $o(1)$ version follows by setting $\delta_{L,K} := C \epsilon_{\text{sub}} / (\gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}})$. $\square$

B.3 Proof Overview

We use the notation of the seam-amortized contraction statement. For a fixed calibration sample $x \in \mathcal{X}$, write $e_b := \hat{X}_b - X_b^*$ for the activation mismatch at depth b , and

$$m_i := \|\hat{X}_{i+1} - X_{i+1}^*\|_F = \|e_{i+1}\|_F, \quad i \in \{-1, 0, \dots, L-2\}, \quad (30)$$

for the midpoint residual of pair $(i, i+1)$, which equals the activation mismatch at depth $i+1$. We adopt $i = -1$ as a formal boundary index corresponding to the common calibration input, so that $m_{-1} = \|\hat{X}_0 - X_0^*\|_F = 0$ since both schedules start from $\hat{X}_0 = X_0^*$; this zero initial condition is used as the base case in Lemma B.9 below. We use the shorthand $\tau := \gamma\rho$ throughout and assume, as in Theorem B.5, that $\tau < 1$. Whenever the same symbol could arise at several moments of the algorithm, it is tied to the specific schedule and time stated in the surrounding sentence; superscripts such as “seq”, “int”, “pre”, “post”, and “out” are used only where the distinction is needed for an inequality. All statements below are first pathwise for this fixed calibration sample, and are averaged only in Section B.8.

The structure of the proof is the following.

1. Section B.4 (Lemma B.9): a uniform one-step recursion on m_i that holds under both schedules for any pair receiving a single CBQ pass.
2. Section B.5 (Lemma B.10): a formal record of the structural fact that every chunk seam is visited by the inner refinement of two consecutive chunks.
3. Section B.6 (Lemma B.11): isolates the contribution of a single chunk seam, where ICBQ’s two-pass certificate yields one extra factor γ relative to a single pass.
4. Section B.7 (Lemma B.13 and Propositions B.14–B.15): chains the per-pair and seam recurrences along the depth axis, producing closed-form upper bounds $\mathcal{U}_i, \mathcal{V}_i$ on the midpoint residual at each pair.
5. Section B.8: extracts the final-prefix identity for $\hat{X}_{L-1}$ via Lemma B.17 (final-visit-time monotonicity) and Lemma B.18 (final-prefix invariance), and then assembles Theorem B.5.
6. Section B.9 (Proposition B.20): the per-seam factor γ^S is tight at the recurrence level.
7. Section B.10 (Proposition B.21): extension to the locally expansive regime $\rho > 1$ under $\tau < 1$.

B.4 Per-pair recursion

Lemma B.9 (Per-pair recursion). *Fix a schedule $\mathcal{S} \in \{\mathcal{S}_{\text{seq}}, \mathcal{S}_{\text{int}}\}$. For every pair $(i, i+1)$ that receives a single CBQ pass under $\mathcal{S}$, the midpoint residual obeys*

$$m_i^{\text{out}} \leq \gamma\rho m_{i-1}^{\text{out}} + (1 + \gamma) \epsilon_{\text{sub}}, \quad i \geq 0, \quad (31)$$

with $m_{-1}^{\text{out}} := 0$ (both schedules start from the common calibration input $\hat{X}_0 = X_0^*$). The inequality holds pathwise for every calibration sample.

Proof. Fix an arbitrary calibration sample $n \in \{1, \dots, N\}$; we suppress the superscript (n) in what follows since every quantity below is evaluated on this single sample. Let $\hat{X}_i$ and X_i^* denote the student and teacher activations at depth i in $\mathcal{S}$ immediately before pair $(i, i+1)$ is processed. By Assumption B.1, every student activation arising in either schedule at depth i lies in $\mathcal{T}_i$, as does X_i^* ; hence $\hat{X}_i, X_i^* \in \mathcal{T}_i$. By Eq. (30), $m_{i-1}^{\text{out}} = \|\hat{X}_i - X_i^*\|_F$. Let θ_i^{pre} and $\theta_{i+1}^{\text{pre}}$ denote the parameters of blocks i and $i+1$ at that moment (obtained from the prefit step when present, the inner quantizer, and any earlier CBQ updates that $\mathcal{S}$ has performed on those blocks). Both states arise at an intermediate step of $\mathcal{S}$, so $\theta_i^{\text{pre}} \in \Theta_i$ and $\theta_{i+1}^{\text{pre}} \in \Theta_{i+1}$ by the construction of Θ_i, Θ_{i+1} in Assumption B.2. Thus the CBQ call currently being processed is one of the actual calls covered by the midpoint contraction in Assumption B.2: its pre-pass states lie in $\Theta_i \times \Theta_{i+1}$ and its sample input $\hat{X}_i$ lies in $\mathcal{T}_i$.

Bound on the pre-pass residual. The pre-pass midpoint residual of pair $(i, i+1)$ is

$$r^{\text{in}} = \|f_i^{\theta^{\text{pre}}}(\hat{X}_i) - X_{i+1}^*\|_F = \|f_i^{\theta^{\text{pre}}}(\hat{X}_i) - f_i^*(X_i^*)\|_F,$$

where we used $X_{i+1}^* = f_i^*(X_i^*)$. By the triangle inequality,

$$\begin{aligned} r^{\text{in}} &\leq \underbrace{\|f_i^{\theta^{\text{pre}}}(\hat{X}_i) - f_i^*(\hat{X}_i)\|_F}_{\leq \epsilon_{\text{sub}} \text{ by Eq. (19)}} + \underbrace{\|f_i^*(\hat{X}_i) - f_i^*(X_i^*)\|_F}_{\leq \rho \|\hat{X}_i - X_i^*\|_F = \rho m_{i-1}^{\text{out}} \text{ by Asm. B.1}} \\ &\leq \rho m_{i-1}^{\text{out}} + \epsilon_{\text{sub}}. \end{aligned} \quad (32)$$

where the first bound applies the local envelope in Eq. (19) (valid because $\theta_i^{\text{pre}} \in \Theta_i$ and $\hat{X}_i \in \mathcal{T}_i$) and the second uses Assumption B.1 (valid because $\hat{X}_i, X_i^* \in \mathcal{T}_i$).

Applying the midpoint contraction. Using Eq. (21) on the single pass on pair $(i, i+1)$,

$$m_i^{\text{out}} = r^{\text{out}} \leq \gamma r^{\text{in}} + \epsilon_{\text{sub}} \leq \gamma(\rho m_{i-1}^{\text{out}} + \epsilon_{\text{sub}}) + \epsilon_{\text{sub}} = \gamma \rho m_{i-1}^{\text{out}} + (1 + \gamma) \epsilon_{\text{sub}}, \quad (33)$$

where the first equality uses $m_i^{\text{out}} = \|\hat{X}_{i+1} - X_{i+1}^*\|_F = \|f_i^{\theta^{\text{post}}}(\hat{X}_i) - X_{i+1}^*\|_F = r^{\text{out}}$ (since the student advance from $\hat{X}_i$ to $\hat{X}_{i+1}$ uses the post-update block i). This is exactly Eq. (31). The base case $m_{-1}^{\text{out}} = 0$ holds because both schedules share the common calibration input $\hat{X}_0 = X_0^*$; the convention $m_{-1}^{\text{out}} = \|\hat{X}_0 - X_0^*\|_F$ follows from Eq. (30) with $i = -1$. $\square$

B.5 Structural fact: every seam pair is visited twice

Before isolating the seam contraction quantitatively, we first record, as a separate lemma, the structural “visited twice” property of the ICBQ schedule.

Lemma B.10 (Seam structural property). *Let $L \geq 2$ and $1 \leq K \leq L$, with $S = S(L, K) = \lceil L/K \rceil - 1$ the number of chunk seams. For every $c \in \{1, \dots, S\}$:*

- (a) *the seam index $cK-1$ is a valid pair index, i.e. $cK-1 \leq L-2$;*
- (b) *the boundary pair $(cK-1, cK)$ is the last pair processed in chunk c ’s inner refinement, and the first pair processed in chunk $c+1$ ’s inner refinement;*
- (c) *under ICBQ, the boundary pair $(cK-1, cK)$ is therefore optimized exactly twice.*

Proof. Validity (a). If $S = 0$ there is nothing to prove; assume henceforth $S \geq 1$. Write $L = qK + r$ with $q = \lfloor L/K \rfloor$ and $0 \leq r < K$. If $r = 0$ then $\lceil L/K \rceil = q$, $S = q-1 \geq 1$ (so $q \geq 2$), and $SK = (q-1)K = L-K \leq L-1$ since $K \geq 1$. If $r \geq 1$ then $\lceil L/K \rceil = q+1$, $S = q \geq 1$, and $SK = qK = L-r \leq L-1$ since $r \geq 1$. In both cases $SK \leq L-1$, hence for every $c \in \{1, \dots, S\}$ the seam index satisfies $cK-1 \leq SK-1 \leq L-2$, which proves (a).

Pair-loop placement (b). For chunk $c \in \{1, \dots, S\}$, the chunk-pointer satisfies $b_{\text{cs}} = (c-1)K$ at the start of chunk c , and the chunk closes when the outer driver reaches $b = cK-1$ (i.e. $b+1 - b_{\text{cs}} = K$). By the chunk-closure window rule, the window range at this closure is $[w_{\min}, w_{\max}) = [\max(0, (c-1)K-1), \min(cK, L-1))$. Since $cK \leq L-1$ by (a), $w_{\max} = cK$, hence the largest valid pair index in chunk c ’s pair loop is $w_{\max} - 1 = cK-1$, so the seam pair $(cK-1, cK)$ is the last pair processed in chunk c ’s inner refinement. Conversely, chunk $c+1$ starts with $b_{\text{cs}} = cK$, so its window range begins at $w_{\min} = \max(0, cK-1) = cK-1$ (since $cK \geq K \geq 1$), making $(cK-1, cK)$ the first pair processed in chunk $c+1$ ’s inner refinement.

Multiplicity (c). By (b), the seam pair $(cK-1, cK)$ is processed once at the end of chunk c ’s pair loop and once at the start of chunk $c+1$ ’s pair loop, hence optimized exactly twice in total. $\square$

B.6 Single-seam lemma

The following lemma isolates the contribution of a single chunk seam where ICBQ and sequential CBQ differ.

Lemma B.11 (Single-seam contraction). *Fix $c \in \{1, \dots, S\}$ and consider the seam pair $(cK-1, cK)$. Let $\hat{X}_{cK-1} \in \mathcal{T}_{cK-1}$ be the student activation at depth $cK-1$ immediately before the first CBQ pass on this pair, and let $(\theta_{cK-1}^{\text{pre}}, \theta_{cK}^{\text{pre}}) \in \Theta_{cK-1} \times \Theta_{cK}$ be the pre-first-pass parameters of blocks $cK-1$ and cK . For the actual ICBQ seam state considered in part (ii), the second component is the fp16 teacher copy, $\theta_{cK}^{\text{pre}} = \theta_{cK}^*$; part (i) records the contraction certified for this first pass. Let*

$$\tilde{m} := \|f_{cK-1}^{\theta_{cK-1}^{\text{pre}}}(\hat{X}_{cK-1}) - X_{cK}^*\|_F. \quad (34)$$

Then:

- (i) (One certified pass.) *The first CBQ pass on $(cK-1, cK)$ from this actual seam state leaves a post-pair midpoint residual*

$$m_{cK-1}^{(1), \text{out}} \leq \gamma \tilde{m} + \epsilon_{\text{sub}}. \quad (35)$$

- (ii) (Interleaved.) *Under ICBQ, the seam pair $(cK-1, cK)$ is visited twice (Lemma B.10): pass 1 is the last pair in chunk c 's inner refinement, and pass 2 is the first pair in chunk $c+1$'s inner refinement. Between the two passes the outer driver executes the end-of-chunk re-roll at the close of chunk c , the prefit/inner-quantizer/one-layer-advance sequence on blocks $cK, \dots, \min\{(c+1)K-1, L-1\}$, and the in-pass re-roll opening chunk $c+1$'s refinement; none of these operations modifies the parameters of blocks $0, \dots, cK-1$. In this setting the post-seam midpoint residual satisfies*

$$m_{cK-1}^{\text{int}, \text{out}} \leq \gamma^2 \tilde{m} + (1 + \gamma) \epsilon_{\text{sub}}. \quad (36)$$

Proof. Claim (i). The pre-pass state $(\theta_{cK-1}^{\text{pre}}, \theta_{cK}^{\text{pre}})$ of the first seam pass is admissible by hypothesis, i.e. it lies in $\Theta_{cK-1} \times \Theta_{cK}$; the student activation $\hat{X}_{cK-1}$ presented to the pair lies in $\mathcal{T}_{cK-1}$. Since this is an actual CBQ call of the interleaved schedule, the midpoint contraction in Eq. (21) applies directly and yields Eq. (35).

Claim (ii). Write $\theta_{cK-1}^{(1), \text{post}}$ and $\theta_{cK}^{(1), \text{post}}$ for the states of blocks $cK-1$ and cK after pass 1; Adam updates both block parameters jointly during the look-ahead pass, despite the pre-pass state of block cK being the fp16 teacher copy θ_{cK}^* , which lies in Θ_{cK} by the definition preceding Assumption B.2. Since the pre-pass pair $(\theta_{cK-1}^{\text{pre}}, \theta_{cK}^{\text{pre}}) = (\theta_{cK-1}^{\text{pre}}, \theta_{cK}^*)$ is admissible and $\hat{X}_{cK-1} \in \mathcal{T}_{cK-1}$, Assumption B.2 applied to the actual pass 1 provides $(\theta_{cK-1}^{(1), \text{post}}, \theta_{cK}^{(1), \text{post}}) \in \Theta_{cK-1} \times \Theta_{cK}$ together with the pathwise midpoint contraction

$$r^{(1)} := \|f_{cK-1}^{\theta_{cK-1}^{(1), \text{post}}}(\hat{X}_{cK-1}) - X_{cK}^*\|_F \leq \gamma \tilde{m} + \epsilon_{\text{sub}}. \quad (37)$$

Observe that $r^{(1)}$ depends on block cK only through the teacher activation $X_{cK}^* = f_{cK-1}^*(X_{cK-1}^*)$ and not through $\theta_{cK}^{(1), \text{post}}$.

We now show that the pre-pass-2 midpoint residual equals $r^{(1)}$. Let $\hat{X}_{cK-1}^{(2)}$ denote the student activation at depth $cK-1$ presented to pass 2, and let $\theta_{cK-1}^{(2), \text{pre}}$ denote the pre-pass-2 state of block $cK-1$. It suffices to establish the two identities

$$\hat{X}_{cK-1}^{(2)} = \hat{X}_{cK-1}, \quad \theta_{cK-1}^{(2), \text{pre}} = \theta_{cK-1}^{(1), \text{post}}. \quad (38)$$

Prefix invariance. By Lemma B.10(b), pass 1 is the *last* pair processed in chunk c 's inner refinement (its index equals $w_{\text{max}} - 1 = cK-1$), and pass 2 is the *first* pair processed in chunk $c+1$'s inner refinement (its index equals $w_{\text{min}} = cK-1$ for the next chunk). We claim that every parameter $\theta_{b''}$ with $b'' \in \{0, \dots, cK-2\}$ takes the *same* value at the start of pass 1 (when chunk c 's pair loop begins processing pair $(cK-1, cK)$) and at the start of pass 2; denote this common value $\theta_{b''}^{(1)} \in \Theta_{b''}$. Indeed, pass 1 is a two-block CBQ subproblem on $(cK-1, cK)$, and its rollback-safeguarded Adam update modifies *only* θ_{cK-1} and θ_{cK} (no other parameter vector enters the CBQ loss $\mathcal{L}_{\text{cbq}}$ for this pair, and Algorithm 2's advance step at line 5 is a read-only evaluation that does not update θ). Hence $\theta_{b''}$ with $b'' \leq cK-2$ is unchanged by pass 1, so its pre-pass-1 and end-of-pass-1 values agree.

Between the end of pass 1 and the start of pass 2, the outer driver executes exactly

- (a) the end-of-chunk activation re-roll at the close of chunk c ;
- (b) for each $b' \in \{cK, cK+1, \dots, \min\{(c+1)K-1, L-1\}\}$, the prefit step (when present), the inner quantizer Q on $\theta_{b'}$, and a one-layer advance of the student and teacher streams through the updated block b' ;
- (c) the in-pass re-roll that opens chunk $c+1$'s inner refinement.

Operations (a) and (c) recompute activations only, not parameters. Operation (b) modifies $\theta_{b'}$ only for $b' \geq cK$. Hence every parameter $\theta_{b''}$ with $b'' \in \{0, \dots, cK-1\}$ is unchanged between the end of pass 1 and the start of pass 2. Combining this with the invariance under pass 1 for $b'' \leq cK-2$,

$$\theta_{b''}^{\text{start of pass 1}} = \theta_{b''}^{\text{end of pass 1}} = \theta_{b''}^{\text{start of pass 2}} \quad \text{for } b'' \in \{0, \dots, cK-2\}, \quad (39)$$

which we take as the definition of $\theta_{b''}^{(1)}$; membership $\theta_{b''}^{(1)} \in \Theta_{b''}$ follows from the construction of $\Theta_{b''}$ in Assumption B.2. For $b'' = cK-1$, operations (a)–(c) leave θ_{cK-1} unchanged, so $\theta_{cK-1}^{\text{end of pass 1}} = \theta_{cK-1}^{\text{start of pass 2}} = \theta_{cK-1}^{(1), \text{post}} = \theta_{cK-1}^{(2), \text{pre}}$ by the very definition of these symbols.

Second identity in Eq. (38). The chain $\theta_{cK-1}^{\text{end of pass 1}} = \theta_{cK-1}^{\text{start of pass 2}}$ just established provides $\theta_{cK-1}^{(2), \text{pre}} = \theta_{cK-1}^{(1), \text{post}}$; in particular $\theta_{cK-1}^{(2), \text{pre}} \in \Theta_{cK-1}$.

First identity in Eq. (38). For $q \in \{1, 2\}$, let $\Phi^{(q)}$ denote the prefix map from depth 0 to depth $cK-1$ using the parameters at the start of pass q :

$$\Phi^{(q)} := \begin{cases} \text{Id}, & cK-1 = 0, \\ f_{cK-2}^{\theta_{cK-2}^{\text{start of pass } q}} \circ \dots \circ f_0^{\theta_0^{\text{start of pass } q}}, & cK-1 \geq 1. \end{cases} \quad (40)$$

By Assumption B.4, the in-pass re-roll at the start of pass 2 produces $\hat{X}_{cK-1}^{(2)} = \Phi^{(2)}(X_0)$ on the stored calibration inputs. Inside chunk c 's inner-refinement pair loop, the same prefix map is applied incrementally before processing pair $(cK-1, cK)$, so $\hat{X}_{cK-1} = \Phi^{(1)}(X_0)$. By Eq. (39), the two prefix maps use *identical* parameters $\theta_{b''}^{(1)}$ for every $b'' \in \{0, \dots, cK-2\}$; when this index set is empty, both maps are Id. Hence $\hat{X}_{cK-1}^{(2)} = \hat{X}_{cK-1}$, which establishes the first identity. In particular, $\hat{X}_{cK-1}^{(2)} \in \mathcal{T}_{cK-1}$ since $\hat{X}_{cK-1} \in \mathcal{T}_{cK-1}$ by hypothesis.

Applying the contraction to pass 2. We first record the pre-pass-2 state of block cK . Operation (b) in the interval between pass 1 and pass 2 applies to block cK (the $b' = cK$ step) either the prefit step followed by the inner quantizer Q (when $Q = \text{DBF}$) or Q alone (when $Q = \text{GPTQ}$); in both cases the resulting post-inner-quantizer state lies in Θ_{cK} by the definition preceding Assumption B.2. Subsequent operations in the interval (the in-pass re-roll (c), the remaining prefit/inner-quantizer steps for $b' \in \{cK+1, \dots, \min\{(c+1)K-1, L-1\}\}$ if any, and the earlier end-of-chunk re-roll (a)) do not touch θ_{cK} . Hence $\theta_{cK}^{(2), \text{pre}}$ equals this post-inner-quantizer state and lies in Θ_{cK} . Combined with $\theta_{cK-1}^{(2), \text{pre}} \in \Theta_{cK-1}$ and $\hat{X}_{cK-1}^{(2)} \in \mathcal{T}_{cK-1}$, this actual pass 2 is covered by the midpoint-contraction certificate Eq. (21). The teacher activation X_{cK}^* is unchanged between the two passes because the teacher blocks are fixed by Assumption B.1 and are reproduced verbatim by the teacher-side re-rolls of Assumption B.4. Combining the two identities in Eq. (38),

$$r^{(2), \text{in}} := \|f_{cK-1}^{\theta_{cK-1}^{(2), \text{pre}}}(\hat{X}_{cK-1}^{(2)}) - X_{cK}^*\|_F = \|f_{cK-1}^{\theta_{cK-1}^{(1), \text{post}}}(\hat{X}_{cK-1}) - X_{cK}^*\|_F = r^{(1)}. \quad (41)$$

Crucially, $r^{(2), \text{in}}$ depends on block cK only through the teacher target X_{cK}^* and not through $\theta_{cK}^{(2), \text{pre}}$; the transition $\theta_{cK}^{(1), \text{post}} \mapsto \theta_{cK}^{(2), \text{pre}}$ induced by the intervening prefit/inner-quantizer steps therefore does not enter the midpoint residual. This transition does enter the subsequent Adam trajectory of pass 2, since the aggregate CBQ loss $\mathcal{L}_{\text{cbq}}$ depends on both block parameters; Assumption B.2 accommodates this by certifying the post-pass midpoint contraction for the actual admissible pre-pass pair $(\theta_{cK-1}^{(2), \text{pre}}, \theta_{cK}^{(2), \text{pre}}) \in \Theta_{cK-1} \times \Theta_{cK}$ produced by operation (b). Applying Eq. (21) to pass 2 with input residual $r^{(2), \text{in}} = r^{(1)}$ and combining with Eq. (37) yields

$$m_{cK-1}^{\text{int, out}} \leq \gamma r^{(2), \text{in}} + \epsilon_{\text{sub}} = \gamma r^{(1)} + \epsilon_{\text{sub}} \leq \gamma^2 \tilde{m} + (1 + \gamma) \epsilon_{\text{sub}}, \quad (42)$$

which is Eq. (36). $\square$

Remark B.12. The ratio of ICBQ’s post-seam residual to the sequential one is γ in the leading term, at the cost of a single additional $\gamma \epsilon_{\text{sub}}$ in the residual term. The leading-term gain is therefore visible whenever this additive term is small relative to the incoming seam residual.

B.7 Chain along the depth axis

We now chain Lemmas B.9 and B.11 along the full depth $0, \dots, L-1$. Let $u_i := m_i^{\text{seq, out}}$ and $v_i := m_i^{\text{int, out}}$ denote the post-pair midpoint residuals of the two schedules at pair $i = 0, \dots, L-2$. Write

$$\tau := \gamma\rho, \quad R := (1 + \gamma) \epsilon_{\text{sub}}, \quad \tilde{R} := (1 + \gamma + \gamma^2) \epsilon_{\text{sub}}, \quad (43)$$

and let $s(i) := |\{c \in \{1, \dots, S\} : cK-1 \leq i\}|$ be the number of chunk seams whose index is at most i . Both schedules share the initial condition $u_{-1} = v_{-1} = 0$ (the calibration-sample-matched inputs) and satisfy the recurrences in the following lemma; the remaining work of this subsection is to unroll them.

Lemma B.13 (Per-pair recurrences). *For every $i \in \{0, \dots, L-2\}$,*

- (S) $u_i \leq \tau u_{i-1} + R$ uniformly over i (sequential CBQ);
- (I) $v_i \leq \tau v_{i-1} + R$ if i is a non-seam pair, and $v_i \leq \gamma\tau v_{i-1} + \tilde{R}$ if $i = cK-1$ is a seam pair (ICBQ).

Proof. Statement (S). Sequential CBQ processes every pair exactly once, so Lemma B.9 applied to pair $(i, i+1)$ yields $u_i \leq \gamma\rho u_{i-1} + (1 + \gamma) \epsilon_{\text{sub}} = \tau u_{i-1} + R$.

Non-seam case of (I). For a non-seam pair i that receives a single CBQ pass, Lemma B.9 gives $v_i \leq \tau v_{i-1} + R$, identical to (S).

Seam case of (I). Let $i = cK-1$ be a seam pair. In the interleaved schedule, let $\hat{X}_{cK-1}$ denote the student activation at depth $cK-1$ presented to pass 1, and let $(\theta_{cK-1}^{\text{pre}}, \theta_{cK}^{\text{pre}})$ denote the pre-pass-1 state of blocks $cK-1$ and cK . Both components of this pair arise at an intermediate step of $\mathcal{S}_{\text{int}}$: $\theta_{cK-1}^{\text{pre}}$ is either the state left by the previous pair update in chunk c ’s inner refinement (when $cK-1 \geq 1$ and chunk c ’s pair loop has processed at least one pair before the seam) or the post-inner-quantizer state of block $cK-1$ (when the seam is the first pair processed in chunk c , e.g. for $c = 1, K = 1$); both belong to Θ_{cK-1} by the definition preceding Assumption B.2. The other component $\theta_{cK}^{\text{pre}} = \theta_{cK}^*$ is the fp16 teacher copy (block cK has not yet been processed by the prefit or inner-quantizer step at the start of pass 1), which belongs to Θ_{cK} by the same definition. Hence $(\theta_{cK-1}^{\text{pre}}, \theta_{cK}^{\text{pre}}) \in \Theta_{cK-1} \times \Theta_{cK}$, verifying the hypothesis of Lemma B.11. By the construction of $\mathcal{T}_{cK-1}$ in Assumption B.1, $\hat{X}_{cK-1}, X_{cK-1}^* \in \mathcal{T}_{cK-1}$.

We claim that $\|\hat{X}_{cK-1} - X_{cK-1}^*\|_F = v_{i-1}$ (with the convention $v_{-1} = 0$) in both of the following cases:

- If $cK-1 \geq 1$ (which includes $c \geq 2$, and $c = 1$ with $K \geq 2$), chunk c ’s inner-refinement loop processes pair $(cK-2, cK-1)$ before pair $(cK-1, cK)$ —the latter is the last pair in chunk c ’s window—so $\hat{X}_{cK-1}$ equals the post-pair- $(cK-2)$ student activation, and $\|\hat{X}_{cK-1} - X_{cK-1}^*\|_F = m_{cK-2}^{\text{int, out}} = v_{i-1}$ by Eq. (30).
- If $cK-1 = 0$ (which happens only for $c = 1, K = 1$), the seam pair $(0, 1)$ is the only pair in chunk 1’s window, and the in-pass re-roll (Assumption B.4) gives $\hat{X}_0 = X_0^*$; hence $\|\hat{X}_0 - X_0^*\|_F = 0 = v_{-1}$ by the boundary convention.

In either case $\|\hat{X}_{cK-1} - X_{cK-1}^*\|_F = v_{i-1}$.

Let $\tilde{m} = \|f_{cK-1}^{\theta_{cK-1}^{\text{pre}}}(\hat{X}_{cK-1}) - X_{cK}^*\|_F$ denote the pre-seam midpoint residual presented to pass 1 (in the notation of Lemma B.11). By the same triangle-inequality derivation as in Eq. (32) (local envelope in Eq. (19) applied with $\theta_{cK-1} = \theta_{cK-1}^{\text{pre}}, z = \hat{X}_{cK-1}$, and Assumption B.1 applied with $x = \hat{X}_{cK-1}, y = X_{cK-1}^*$),

$$\tilde{m} \leq \rho \|\hat{X}_{cK-1} - X_{cK-1}^*\|_F + \epsilon_{\text{sub}} = \rho v_{i-1} + \epsilon_{\text{sub}}. \quad (44)$$

Applying Lemma B.11(ii) (the two-pass seam contraction $v_i \leq \gamma^2 \tilde{m} + (1 + \gamma)\epsilon_{\text{sub}}$) with the hypotheses just verified, and combining with Eq. (44) and $\tau = \gamma\rho$,

$$\begin{aligned} v_i &\leq \gamma^2 \tilde{m} + (1 + \gamma)\epsilon_{\text{sub}} \\ &\leq \gamma^2 (\rho v_{i-1} + \epsilon_{\text{sub}}) + (1 + \gamma)\epsilon_{\text{sub}} \\ &= \gamma\rho \cdot \gamma v_{i-1} + (\gamma^2 + 1 + \gamma)\epsilon_{\text{sub}} \\ &= \gamma\tau v_{i-1} + \tilde{R}, \end{aligned}$$

where the last equality uses $\tilde{R} = (1 + \gamma + \gamma^2)\epsilon_{\text{sub}}$. $\square$

Unrolling Lemma B.13(S) from $u_{-1} = 0$ yields a closed-form sequential upper bound $\mathcal{U}_i$; unrolling Lemma B.13(I) provides a closed-form interleaved upper bound $\mathcal{V}_i$ in which a contribution originating from pair k is multiplied by an additional factor γ for each subsequent seam.

Proposition B.14 (Pair-level unrolling). *Under Assumptions B.1, B.2, and B.4 and $\tau < 1$, for every $i \in \{0, \dots, L-2\}$ the midpoint residuals satisfy $u_i \leq \mathcal{U}_i$ and $v_i \leq \mathcal{V}_i$ with the deterministic upper bounds*

$$\mathcal{U}_i := R \sum_{k=0}^i \tau^{i-k} = R \frac{1 - \tau^{i+1}}{1 - \tau}, \quad (45)$$

$$\mathcal{V}_i := \tilde{R} \sum_{k=0}^i \tau^{i-k} \gamma^{s(i)-s(k)} \leq \tilde{R} \frac{1 - \tau^{i+1}}{1 - \tau}. \quad (46)$$

Moreover the two bounds admit the direct comparison

$$\mathcal{V}_i \leq \gamma^{s(i)} \mathcal{U}_i + (\tilde{R} - \gamma^{s(i)} R) \frac{1 - \tau^{i+1}}{1 - \tau}, \quad (47)$$

where the factor $\gamma^{s(i)}$ quantifies the cumulative seam-amortized gain in the propagated term and $(\tilde{R} - \gamma^{s(i)} R) \geq 0$ upper-bounds the additive terms not carrying this full factor.

Proof. Proof of Eq. (45). From $u_{-1} = 0$, Lemma B.13(S) gives, by a one-step induction, the base $u_0 \leq \tau \cdot 0 + R = R = \mathcal{U}_0$ and the step $u_i \leq \tau u_{i-1} + R \leq \tau \mathcal{U}_{i-1} + R = R(\tau \sum_{k=0}^{i-1} \tau^{i-1-k} + 1) = R \sum_{k=0}^i \tau^{i-k} = \mathcal{U}_i$.

Proof of Eq. (46). Define, for $\ell \in \{0, \dots, L-2\}$,

$$\alpha_\ell := \begin{cases} \tau & \text{if } \ell \text{ is non-seam,} \\ \gamma\tau & \text{if } \ell = cK-1 \text{ is seam,} \end{cases} \quad \beta_\ell := \begin{cases} R & \text{if } \ell \text{ is non-seam,} \\ \tilde{R} & \text{if } \ell = cK-1 \text{ is seam.} \end{cases} \quad (48)$$

Both α_ℓ, β_ℓ are nonnegative, and $0 \leq \beta_\ell \leq \tilde{R}$ (since $\tilde{R} = R + \gamma^2 \epsilon_{\text{sub}} \geq R$). With the convention that an empty product is 1, induction from $v_{-1} = 0$ using Lemma B.13(I) gives

$$v_i \leq \sum_{k=0}^i \beta_k \prod_{\ell=k+1}^i \alpha_\ell \leq \tilde{R} \sum_{k=0}^i \prod_{\ell=k+1}^i \alpha_\ell, \quad (49)$$

because the base case is $v_0 \leq \alpha_0 v_{-1} + \beta_0 = \beta_0$, and the induction step follows from

$$v_i \leq \alpha_i v_{i-1} + \beta_i \leq \sum_{k=0}^{i-1} \beta_k \prod_{\ell=k+1}^i \alpha_\ell + \beta_i. \quad (50)$$

The second inequality in Eq. (49) uses $\beta_k \leq \tilde{R}$ and the nonnegativity of the product. The product $\prod_{\ell=k+1}^i \alpha_\ell$ picks up a factor τ at every $\ell \in \{k+1, \dots, i\}$ and an *extra* factor γ at every seam in that range. The number of seams in $\{k+1, \dots, i\}$ is $s(i) - s(k)$ by definition of $s(\cdot)$, so

$$\prod_{\ell=k+1}^i \alpha_\ell = \tau^{i-k} \gamma^{s(i)-s(k)}. \quad (51)$$

The product identity therefore gives the interleaved closed form

$$v_i \leq \mathcal{V}_i = \tilde{R} \sum_{k=0}^i \tau^{i-k} \gamma^{s(i)-s(k)}.$$

Its geometric upper bound follows from $\gamma^{s(i)-s(k)} \leq 1$, valid because $s(i) \geq s(k)$ and $\gamma \in (0, 1)$.

Direct comparison. The quantities $\mathcal{U}_i, \mathcal{V}_i$ are deterministic once $\tau, \gamma, R, \tilde{R}$ are fixed. Comparing their closed forms and pulling the geometric factor $\gamma^{s(i)}$ out of $\mathcal{U}_i$,

$$\mathcal{V}_i - \gamma^{s(i)} \mathcal{U}_i = \sum_{k=0}^i \tau^{i-k} \left[\tilde{R} \gamma^{s(i)-s(k)} - \gamma^{s(i)} R \right]. \quad (52)$$

For each term, $\gamma^{s(i)-s(k)} \leq 1$ and $\tilde{R} \geq 0$ imply $\tilde{R} \gamma^{s(i)-s(k)} \leq \tilde{R}$, so

$$\mathcal{V}_i - \gamma^{s(i)} \mathcal{U}_i \leq \sum_{k=0}^i \tau^{i-k} (\tilde{R} - \gamma^{s(i)} R) = (\tilde{R} - \gamma^{s(i)} R) \frac{1 - \tau^{i+1}}{1 - \tau}. \quad (53)$$

Rearranging yields the stated comparison. The coefficient $\tilde{R} - \gamma^{s(i)} R$ is nonnegative: since $\gamma \in (0, 1)$ and $s(i) \geq 0$, $\gamma^{s(i)} \leq 1$, and thus $\gamma^{s(i)} R \leq R \leq \tilde{R}$. Hence the comparison is a nontrivial upper bound of $\mathcal{V}_i$, and the two components have the claimed interpretation: $\gamma^{s(i)} \mathcal{U}_i$ is the propagated contribution amortized by $s(i)$ seams, and $(\tilde{R} - \gamma^{s(i)} R)(1 - \tau^{i+1})/(1 - \tau)$ collects the additive terms that do not carry the full $\gamma^{s(i)}$ factor. $\square$

The bound in Eq. (47) is convenient for matching the sequential envelope $\mathcal{U}_i$, but it overestimates the residual contribution of non-seam pairs by replacing each β_k by the seam-only upper bound $\tilde{R}$. The next proposition records a seam-aware refinement that splits the contribution of seam and non-seam pairs explicitly; it is used only to justify Remark B.6 and is not invoked elsewhere in the proof.

Proposition B.15 (Seam-aware sharper comparison). *Let $\mathcal{J}_i := \{k \in \{0, \dots, i\} : k = cK - 1 \text{ for some } c \in \{1, \dots, S\}\}$ be the set of seam indices up to pair i . Under the hypotheses of Proposition B.14,*

$$v_i \leq \sum_{k=0}^i (R + \gamma^2 \epsilon_{\text{sub}} \mathbf{1}\{k \in \mathcal{J}_i\}) \tau^{i-k} \gamma^{s(i)-s(k)}. \quad (54)$$

Consequently,

$$v_i \leq \gamma^{s(i)} \mathcal{U}_i + D_i \epsilon_{\text{sub}}, \quad (55)$$

where

$$D_i := (1 + \gamma) \sum_{k=0}^i \tau^{i-k} (\gamma^{s(i)-s(k)} - \gamma^{s(i)}) + \gamma^2 \sum_{k \in \mathcal{J}_i} \tau^{i-k} \gamma^{s(i)-s(k)}. \quad (56)$$

In particular, when $S(L, K) = 0$ (single chunk, no seams), $\mathcal{J}_i = \emptyset$ and $s(i) = s(k) = 0$ for every $k \leq i$, so $D_i = 0$ and $v_i \leq \mathcal{U}_i$ exactly.

Proof. The interleaved forcing coefficient is $\beta_k = R$ at non-seam pairs and $\beta_k = \tilde{R} = R + \gamma^2 \epsilon_{\text{sub}}$ at seam pairs, that is, $\beta_k = R + \gamma^2 \epsilon_{\text{sub}} \mathbf{1}\{k \in \mathcal{J}_i\}$. Keeping this seam-dependent forcing in the exact variation-of-constants expansion

$$v_i \leq \sum_{k=0}^i \beta_k \prod_{\ell=k+1}^i \alpha_\ell$$

(without invoking $\beta_k \leq \tilde{R}$) and using the same product identity gives the seam-aware closed form. To extract the comparison with the sequential envelope, write

$$v_i - \gamma^{s(i)} \mathcal{U}_i = R \sum_{k=0}^i \tau^{i-k} (\gamma^{s(i)-s(k)} - \gamma^{s(i)}) + \gamma^2 \epsilon_{\text{sub}} \sum_{k \in \mathcal{J}_i} \tau^{i-k} \gamma^{s(i)-s(k)},$$

using $R = (1 + \gamma)^{\epsilon_{\text{sub}}}$ and rearranging. Both summands are nonnegative because $s(i) \geq s(k)$ implies $\gamma^{s(i)-s(k)} \geq \gamma^{s(i)}$ (as $\gamma \in (0, 1)$). When $S = 0$, $\mathcal{J}_i = \emptyset$ trivially zeros the second sum, and the first sum vanishes term by term because $s(i) - s(k) = 0$. Hence $D_i = 0$ and the bound becomes $v_i \leq \gamma^{s(i)} \mathcal{U}_i = \mathcal{U}_i$. $\square$

Corollary B.16 (Last-pair upper bounds). *At $i = L-2$ (so $s(L-2) = S$), $u_{L-2} \leq \mathcal{U}_{L-2}$ and $v_{L-2} \leq \mathcal{V}_{L-2}$, with*

$$\mathcal{U}_{L-2} = R \frac{1 - \tau^{L-1}}{1 - \tau}, \quad (57)$$

$$\mathcal{V}_{L-2} \leq \gamma^S \mathcal{U}_{L-2} + (\tilde{R} - \gamma^S R) \frac{1 - \tau^{L-1}}{1 - \tau}. \quad (58)$$

Proof. Apply Proposition B.14 to the last valid pair $i = L-2$. It remains only to check that $s(L-2) = S$. Recall that the seam count is $S = \lceil L/K \rceil - 1$ and that the seams are indexed by $c \in \{1, \dots, S\}$ with seam index $cK-1$. If $S = 0$, there are no seams and $s(L-2) = 0 = S$. Assume henceforth $S \geq 1$. Write $L = qK + r$ with $q = \lfloor L/K \rfloor$ and $0 \leq r < K$. Two cases:

- If $r = 0$, then $\lceil L/K \rceil = q$, so $S = q-1 \geq 1$, i.e. $q \geq 2$, and $SK = (q-1)K = L-K \leq L-1$ because $K \geq 1$.
- If $r \geq 1$, then $\lceil L/K \rceil = q+1$, so $S = q \geq 1$ and $SK = qK = L-r \leq L-1$ because $r \geq 1$.

In both cases $SK \leq L-1$. Hence for every $c \in \{1, \dots, S\}$, the seam index satisfies $cK-1 \leq SK-1 \leq L-2$, and is therefore counted by $s(L-2)$. Conversely, by the definition of S , there are no seams with $c > S$, so no seam index exceeds $SK-1 \leq L-2$. We conclude $s(L-2) = S$. $\square$

B.8 From last-pair residual to depth- L mismatch

We propagate the last-pair midpoint residual through the final block $L-1$ in two formal steps: a monotonicity-of-final-visit-times lemma (Lemma B.17) and a final-prefix-invariance lemma (Lemma B.18). For every $b \in \{0, \dots, L-2\}$, let T_b denote the algorithmic time of the *last* CBQ pass on pair $(b, b+1)$: in Sequential CBQ this is the unique pass on that pair; in ICBQ it is the unique pass if $(b, b+1)$ is a non-seam pair, and the second pass if $(b, b+1)$ is a seam pair (i.e. $b = cK-1$ for some $c \in \{1, \dots, S\}$). We extend by $T_{L-1} := T_{L-2}$, so that block $L-1$'s final state is pinned at T_{L-2} . For each $b \in \{0, \dots, L-1\}$, write θ_b^{post} for the parameter state of block b at time T_b .

Lemma B.17 (Monotonicity of final-visit times and block finalization). *Under the schedules described by Algorithms 1–2,*

- (a) *the times satisfy $T_0 < T_1 < \dots < T_{L-2} = T_{L-1}$;*
- (b) *for every $b \in \{0, \dots, L-1\}$, no operation that modifies θ_b occurs strictly after T_b ; consequently, block b 's state at every time $t \geq T_b$ equals θ_b^{post} , and $\theta_b^{\text{post}} \in \Theta_b$ by Assumption B.2.*

Proof. (a) The outer driver iterates $b = 0, 1, \dots, L-1$ in increasing order, and the inner pair loop processes pairs in increasing order of the first-block index within each chunk (Algorithms 1–2). For non-seam pairs, the unique CBQ pass occurs inside the chunk-closing inner refinement at some outer iteration b' , and its algorithmic time is strictly increasing in the pair index b (later pairs are processed later inside the same pair loop, and later chunks close later than earlier chunks). For a seam pair $(cK-1, cK)$, the second pass occurs at chunk $c+1$'s closure, which is strictly later than chunk c 's closure (where the first pass occurs); the second pass on the seam is itself the first pair processed in chunk $c+1$'s pair loop, so its algorithmic time precedes every other CBQ pass in chunk $c+1$. Combining these observations, $T_0 < T_1 < \dots < T_{L-2}$. The convention $T_{L-1} = T_{L-2}$ is by definition.

(b) The operations that may modify θ_b are

- (A1) the outer-driver prefit/inner-quantizer step on block b , which occurs exactly once at outer iteration b ; call its algorithmic time Q_b ;

- (A2) CBQ updates arising from pairs that include b , namely $(b-1, b)$ (defined only for $b \geq 1$) and $(b, b+1)$ (defined only for $b \leq L-2$); re-rolls recompute activations only and modify no parameter.

For type (A2): the last CBQ pass on $(b, b+1)$ is T_b by definition (for $b \leq L-2$); the last CBQ pass on $(b-1, b)$ is T_{b-1} (for $b \geq 1$). By (a), $T_{b-1} < T_b$ for $1 \leq b \leq L-2$, and $T_{L-2} = T_{L-1}$ for $b = L-1$. Hence no type-(A2) operation occurs strictly after T_b .

For type (A1): Q_b occurs at the end of outer iteration b , which strictly precedes every chunk-closing inner refinement triggered at outer iteration $b' \geq b$. If $b \leq L-2$, the first CBQ pass on pair $(b, b+1)$ occurs inside the chunk closing at some $b' \geq b$, hence strictly after Q_b ; in particular $Q_b < T_b$. If $b = L-1$, then $T_{L-1} = T_{L-2}$ lies within the chunk closing at outer iteration $L-1$ (the final chunk), which is strictly after Q_{L-1} , the quantizer step of outer iteration $L-1$; hence $Q_{L-1} < T_{L-1}$.

Combining the two cases, no operation modifying θ_b occurs strictly after T_b , so block b 's state at every $t \geq T_b$ equals θ_b^{post} . Admissibility $\theta_b^{\text{post}} \in \Theta_b$ follows from the definition preceding Assumption B.2. $\square$

Lemma B.18 (Final-prefix identity for $\hat{X}_{L-1}$). *Under Assumptions B.1, B.2, and B.4 and the schedules described in Algorithms 1–2, for every $b \in \{0, \dots, L-2\}$ the student activation at depth $b+1$ at time T_b (immediately after the advance step following that CBQ pass) satisfies*

$$\hat{X}_{b+1}^{[T_b]} = (f_b^{\theta_b^{\text{post}}} \circ \dots \circ f_0^{\theta_0^{\text{post}}})(X_0). \quad (59)$$

In particular, $\hat{X}_{L-1} = (f_{L-2}^{\theta_{L-2}^{\text{post}}} \circ \dots \circ f_0^{\theta_0^{\text{post}}})(X_0)$, and this value is preserved by any subsequent activation-only re-roll (no parameter is modified after T_{L-1}).

Proof. We argue by induction on $b \in \{0, \dots, L-2\}$.

Base case $b = 0$. By Assumption B.4 applied with $d = 0$ (empty composition), $\hat{X}_0 = X_0$, and the advance at T_0 gives $\hat{X}_1^{[T_0]} = f_0^{\theta_0^{\text{post}}}(\hat{X}_0) = f_0^{\theta_0^{\text{post}}}(X_0)$, which is Eq. (59) at $b = 0$.

Inductive step. Assume Eq. (59) holds at T_{b-1} (with $b \geq 1$). In the interval $(T_{b-1}, T_b]$ the algorithm executes some combination of outer-driver block-level steps on indices $b' \geq b$ (the blocks whose outer-driver quantization has not yet taken place by time T_{b-1}), end-of-chunk and in-pass re-rolls, and CBQ passes on pairs $(i, i+1)$ with $i \geq b$ (since the last CBQ pass on any pair $(i, i+1)$ with $i < b$ is $T_i \leq T_{b-1}$, the right endpoint of the preceding interval). By Lemma B.17, $\theta_{b'}$ for every $b' \leq b-1$ remains pinned at its final state $\theta_{b'}^{\text{post}}$ throughout this interval.

We now track $\hat{X}_b$ during $(T_{b-1}, T_b]$. The only operations that can modify $\hat{X}_b$ are:

- (i) re-rolls (which recompute the student prefix up to some depth $\geq b$);
- (ii) the advance step at line 5 of Algorithm 2 following a CBQ pass on pair $(i, i+1)$ with $i+1 = b$, i.e. pair $(b-1, b)$.

By Lemma B.17, no CBQ pass on pair $(b-1, b)$ occurs strictly after T_{b-1} , so advance steps of type (ii) do not occur in the open interval $(T_{b-1}, T_b]$. Hence $\hat{X}_b$ is modified only by re-rolls in this interval.

By Assumption B.4, any re-roll in the interval produces $\hat{X}_b = (f_{b-1}^{\theta_{b-1}^{\text{current}}} \circ \dots \circ f_0^{\theta_0^{\text{current}}})(X_0)$; since $\theta_{b'}^{\text{current}} = \theta_{b'}^{\text{post}}$ for every $b' \leq b-1$ throughout the interval, this equals the *final-prefix* value $P_b := (f_{b-1}^{\theta_{b-1}^{\text{post}}} \circ \dots \circ f_0^{\theta_0^{\text{post}}})(X_0)$. The inductive hypothesis at depth $b-1$ states $\hat{X}_b^{[T_{b-1}]} = P_b$, so at the left endpoint of the interval $\hat{X}_b$ is already at the final-prefix value; between re-rolls, $\hat{X}_b$ is stored and untouched by any other operation, hence frozen at P_b ; every re-roll resets it to P_b . Consequently, $\hat{X}_b^{[t]} = P_b$ for every $t \in (T_{b-1}, T_b]$.

At time T_b , the CBQ pass on pair $(b, b+1)$ sees this $\hat{X}_b$ as its input; immediately after the pass, the advance gives $\hat{X}_{b+1}^{[T_b]} = f_b^{\theta_b^{\text{post}}}(\hat{X}_b) = (f_b^{\theta_b^{\text{post}}} \circ \dots \circ f_0^{\theta_0^{\text{post}}})(X_0)$, which is Eq. (59) at b .

Specialising $b = L-2$ yields $\hat{X}_{L-1} = (f_{L-2}^{\theta_{L-2}^{\text{post}}} \circ \dots \circ f_0^{\theta_0^{\text{post}}})(X_0)$. By Lemma B.17(b), no parameter is modified after $T_{L-1} = T_{L-2}$; any later activation-only re-roll therefore recomputes $\hat{X}_{L-1}$ to the same final-prefix value, leaving it unchanged. $\square$

Proof of Theorem B.5. Pathwise depth- L bound. By Lemma B.18,

$$\hat{X}_{L-1} = (f_{L-2}^{\theta_{L-2}^{\text{post}}} \circ \dots \circ f_0^{\theta_0^{\text{post}}})(X_0) \quad (60)$$

holds at the moment immediately after the CBQ pass at time T_{L-2} , and it is preserved by any subsequent activation-only re-roll. In particular, after the final chunk closes, the end-of-chunk re-roll (or equivalently an explicit final forward pass through the frozen final block) computes $\hat{X}_L = f_{L-1}^{\theta_{L-1}^{\text{post}}}(\hat{X}_{L-1})$. This is an exact application of the final block map by Assumption B.4; no later operation modifies θ_{L-1} by Lemma B.17(b) with $b = L-1$.

Hence $\hat{X}_{L-1} \in \mathcal{T}_{L-1}$ by Assumption B.1, $X_{L-1}^* \in \mathcal{T}_{L-1}$ by construction, and $\|\hat{X}_{L-1} - X_{L-1}^*\|_F = m_{L-2}^{\text{out}}$ by definition in Eq. (30). By the triangle inequality followed by the local envelope in Eq. (19) (applied with $\theta_{L-1} = \theta_{L-1}^{\text{post}} \in \Theta_{L-1}$ and $z = \hat{X}_{L-1} \in \mathcal{T}_{L-1}$) and Assumption B.1,

$$\begin{aligned} \|e_L\|_F &= \|f_{L-1}^{\theta_{L-1}^{\text{post}}}(\hat{X}_{L-1}) - f_{L-1}^*(X_{L-1}^*)\|_F \\ &\leq \|f_{L-1}^{\theta_{L-1}^{\text{post}}}(\hat{X}_{L-1}) - f_{L-1}^*(\hat{X}_{L-1})\|_F + \|f_{L-1}^*(\hat{X}_{L-1}) - f_{L-1}^*(X_{L-1}^*)\|_F \\ &\leq \epsilon_{\text{sub}} + \rho \|\hat{X}_{L-1} - X_{L-1}^*\|_F = \rho m_{L-2}^{\text{out}} + \epsilon_{\text{sub}}. \end{aligned} \quad (61)$$

This inequality is pathwise for each calibration sample.

Sequential bound. For the sequential schedule, apply Corollary B.16: the pathwise bound $u_{L-2}^{\text{out},(n)} \leq \mathcal{U}_{L-2}$ holds for every calibration sample $n \in \{1, \dots, N\}$; the right-hand side $\mathcal{U}_{L-2}$ is sample-independent, so this is a uniform pathwise bound. The last-layer Lipschitz inequality then gives, pathwise on each sample, $\|e_L^{\text{seq},(n)}\|_F \leq \rho \mathcal{U}_{L-2} + \epsilon_{\text{sub}}$ for every n . Taking the calibration expectation $\mathbb{E}[\cdot] = \frac{1}{N} \sum_{n=1}^N (\cdot)^{(n)}$ preserves the inequality because the right-hand side is sample-independent, and the sum $\rho \mathcal{U}_{L-2} + \epsilon_{\text{sub}}$ equals $\mathcal{B}_L^{\text{seq}}$ from Eq. (23):

$$\mathbb{E}\|e_L^{\text{seq}}\|_F \leq \rho \mathcal{U}_{L-2} + \epsilon_{\text{sub}} = \mathcal{B}_L^{\text{seq}}. \quad (62)$$

Interleaved bound. For ICBQ, apply Eq. (61) pathwise to each sample n , then use the pathwise bound $v_{L-2}^{\text{out},(n)} \leq \mathcal{V}_{L-2}$ from Corollary B.16 (with $\mathcal{V}_{L-2}$ sample-independent), and finally apply the deterministic bound-level comparison in Eq. (58):

$$\|e_L^{\text{int},(n)}\|_F \leq \rho v_{L-2}^{\text{out},(n)} + \epsilon_{\text{sub}} \leq \rho \mathcal{V}_{L-2} + \epsilon_{\text{sub}} \leq \rho \left[\gamma^S \mathcal{U}_{L-2} + (\tilde{R} - \gamma^S R) \frac{1 - \tau^{L-1}}{1 - \tau} \right] + \epsilon_{\text{sub}}.$$

Using the identity $\epsilon_{\text{sub}} = \gamma^S \epsilon_{\text{sub}} + (1 - \gamma^S) \epsilon_{\text{sub}}$, reshuffling terms by purely algebraic rearrangement, and taking calibration expectations (all right-hand sides are sample-independent, so the inequality is preserved),

$$\begin{aligned} \mathbb{E}\|e_L^{\text{int}}\|_F &\leq \rho \mathcal{V}_{L-2} + \epsilon_{\text{sub}} \\ &\leq \rho \left[\gamma^S \mathcal{U}_{L-2} + (\tilde{R} - \gamma^S R) \frac{1 - \tau^{L-1}}{1 - \tau} \right] + \epsilon_{\text{sub}} \\ &= \gamma^S [\rho \mathcal{U}_{L-2} + \epsilon_{\text{sub}}] + (\tilde{R} - \gamma^S R) \frac{\rho(1 - \tau^{L-1})}{1 - \tau} + (1 - \gamma^S) \epsilon_{\text{sub}} \\ &= \gamma^S \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma) \epsilon_{\text{sub}}, \end{aligned} \quad (63)$$

where the last equality uses the definition in Eq. (23) of $\mathcal{B}_L^{\text{seq}}$ together with the identification

$$C(L, K, \rho, \gamma) \epsilon_{\text{sub}} = (\tilde{R} - \gamma^S R) \frac{\rho(1 - \tau^{L-1})}{1 - \tau} + (1 - \gamma^S) \epsilon_{\text{sub}}. \quad (64)$$

Expanding the residual coefficient $\tilde{R} - \gamma^S R = [(1 + \gamma + \gamma^2) - \gamma^S(1 + \gamma)] \epsilon_{\text{sub}}$ yields the admissible residual constant whenever $\epsilon_{\text{sub}} > 0$. If $\epsilon_{\text{sub}} = 0$, then $R = \tilde{R} = 0$ and $\mathcal{B}_L^{\text{seq}} = 0$, so the preceding

pathwise recurrences give $\mathbb{E}\|e_L^{\text{int}}\|_F = 0$; choosing the same displayed value of $C(L, K, \rho, \gamma)$ leaves $C(L, K, \rho, \gamma) \epsilon_{\text{sub}} = 0$. Hence Eq. (25) is valid in all cases.

Depth-uniform residual constant. Using $\gamma^S \geq 0$,

$$\tilde{R} - \gamma^S R = [(1 + \gamma + \gamma^2) - \gamma^S(1 + \gamma)] \epsilon_{\text{sub}} \leq (1 + \gamma + \gamma^2) \epsilon_{\text{sub}}, \quad (65)$$

together with $(1 - \gamma^S) \leq 1$ and $(1 - \tau^{L-1})/(1 - \tau) \leq 1/(1 - \tau)$ (from $\tau < 1$), we obtain the depth-uniform bound

$$C(L, K, \rho, \gamma) \leq (1 - \gamma^S) + (1 + \gamma + \gamma^2) \frac{\rho(1 - \tau^{L-1})}{1 - \tau} \leq 1 + \frac{(1 + \gamma + \gamma^2)\rho}{1 - \tau}, \quad (66)$$

independent of L and K . Under the standard stability regime $\rho \leq 1$ this simplifies to $C \leq 1 + (1 + \gamma + \gamma^2)/(1 - \tau)$, as stated in Eq. (26); the near-divergent case $\rho > 1$ is handled in Appendix B.10 under the same sufficient condition $\rho < 1/\gamma$ (equivalently $\tau < 1$). Eq. (63) is exactly the statement of Theorem B.5. $\square$

Remark B.19 (On the interpretation of the upper bound). *The derivation above produces the ICBQ bound $\mathcal{B}_L^{\text{int}} := \gamma^S \mathcal{B}_L^{\text{seq}} + C \epsilon_{\text{sub}}$ and the sequential bound $\mathcal{B}_L^{\text{seq}}$ in terms of the same quantities $\rho, \gamma, \epsilon_{\text{sub}}, \tau$. The factor γ^S is attached only to the propagated component of the bound; the additive residual terms are collected in $C \epsilon_{\text{sub}}$ and can dominate when the sequential bound is already small. The actual realized sequential residual $\mathbb{E}\|e_L^{\text{seq}}\|_F$ may be strictly smaller than $\mathcal{B}_L^{\text{seq}}$; thus, Theorem B.5 is a bound-to-bound comparison rather than a realized-residual comparison, which is how Corollary B.8's amortization-gain prediction should be read.*

B.9 Recurrence-level sharpness

Proposition B.20 (Scalar-recurrence saturation and seam-exponent sharpness). *Fix any $L \geq 2$, any chunk length $K \geq 1$ (which determines the seam set $\mathcal{J}_K = \{cK-1 : c = 1, \dots, S\}$), and any constants $R, \tilde{R} \geq 0$, $\tau \in [0, 1)$, $\gamma \in (0, 1]$. Define the saturating scalar recurrences obtained by replacing the inequalities in Lemma B.13 by equalities:*

$$u_i^* = \tau u_{i-1}^* + R, \quad v_i^* = \begin{cases} \tau v_{i-1}^* + R, & i \notin \mathcal{J}_K, \\ \gamma \tau v_{i-1}^* + \tilde{R}, & i \in \mathcal{J}_K, \end{cases} \quad u_{-1}^* = v_{-1}^* = 0. \quad (67)$$

Then:

- (a) (Envelopes.) *Any pathwise sequence (u_i, v_i) with $u_i, v_i \geq 0$ that satisfies the inequalities in Lemma B.13 with the boundary condition $u_{-1} = v_{-1} = 0$ obeys $u_i \leq u_i^*$ and $v_i \leq v_i^*$ for every $i \geq -1$.*
- (b) (Closed form.) *With*

$$\alpha_\ell^* = \begin{cases} \tau, & \ell \notin \mathcal{J}_K, \\ \gamma \tau, & \ell \in \mathcal{J}_K, \end{cases} \quad \beta_\ell^* = \begin{cases} R, & \ell \notin \mathcal{J}_K, \\ \tilde{R}, & \ell \in \mathcal{J}_K, \end{cases}$$

the saturated sequences satisfy

$$u_i^* = R \sum_{k=0}^i \tau^{i-k}, \quad v_i^* = \sum_{k=0}^i \beta_k^* \prod_{\ell=k+1}^i \alpha_\ell^* = \sum_{k=0}^i \beta_k^* \tau^{i-k} \gamma^{s(i)-s(k)}. \quad (68)$$

- (c) (Sharpness of the seam exponent.) *For every $0 \leq k \leq i \leq L-2$, the linear response at index i to an additive impulse injected at index k is exactly*

$$\prod_{\ell=k+1}^i \alpha_\ell^* = \tau^{i-k} \gamma^{s(i)-s(k)}. \quad (69)$$

Consequently, for $\gamma \in (0, 1)$, no recurrence-level argument that uses only Lemma B.13's multipliers can uniformly replace the exponent $s(i) - s(k)$ by any larger exponent; doing so would underbound this exact impulse response. For $\gamma = 1$ the statement is vacuous because seams add no contraction.

Proof. (a) We argue $u_i \leq u_i^*$ and $v_i \leq v_i^*$ by simultaneous induction on $i \geq -1$. The base case $i = -1$ is immediate from $u_{-1} = v_{-1} = 0 = u_{-1}^* = v_{-1}^*$. For the inductive step, fix $i \geq 0$ and assume $u_{i-1} \leq u_{i-1}^*$ and $v_{i-1} \leq v_{i-1}^*$. The maps $x \mapsto \tau x + R$ and $x \mapsto \gamma \tau x + \tilde{R}$ are monotone nondecreasing on $[0, \infty)$. Applying the appropriate map gives $u_i \leq \tau u_{i-1} + R \leq \tau u_{i-1}^* + R = u_i^*$ and, analogously, $v_i \leq v_i^*$, splitting cases according to whether $i \in \mathcal{J}_K$.

(b) Unrolling the saturating scalar recurrences gives the displayed formula for u_i^* . The formula for v_i^* is the standard variation-of-constants expansion for a non-autonomous linear recursion. The product contains one factor τ for each index in $\{k+1, \dots, i\}$ and one additional factor γ for each seam in that same index range. The number of such seams is $s(i) - s(k)$, giving Eq. (68).

(c) Fix $0 \leq k \leq i \leq L-2$ and add a nonnegative impulse a to the forcing term at index k , leaving all other forcing terms and all multipliers unchanged. Let Δ_j be the difference between the perturbed and unperturbed saturated sequences. Then $\Delta_j = 0$ for $j < k$, $\Delta_k = a$, and $\Delta_j = \alpha_j^* \Delta_{j-1}$ for every $j > k$. Hence

$$\Delta_i = a \prod_{\ell=k+1}^i \alpha_\ell^* = a \tau^{i-k} \gamma^{s(i)-s(k)}.$$

This equality is an exact linear response of the recurrence. If, for some $\eta > 0$ and $\gamma \in (0, 1)$, one attempted to replace the factor $\gamma^{s(i)-s(k)}$ by the smaller factor $\gamma^{s(i)-s(k)+\eta}$ uniformly over nonnegative impulses, the bound would fail for this perturbation because $a \tau^{i-k} \gamma^{s(i)-s(k)} > a \tau^{i-k} \gamma^{s(i)-s(k)+\eta}$ whenever $a > 0$ and $\tau > 0$; the case $\tau = 0$ is degenerate: the response is zero for $i > k$, while for $i = k$ the empty product is one and the same contradiction applies. Thus the seam exponent in Eq. (69) is exact at the recurrence level. $\square$

The theorem's $C \epsilon_{\text{sub}}$ residual is the price of keeping these additive contributions explicit at the depth- L bound.

B.10 The $\rho > 1$ (near-divergent) regime

The per-pair recursion in Eq. (31) requires only $\tau = \gamma\rho < 1$ for $C(L, K, \rho, \gamma)$ to be uniformly bounded in depth. When the fp16 blocks are expansive on parts of the empirical calibration trajectories ($\rho > 1$), ICBQ remains stable provided $\gamma\rho < 1$, while the sequential recursion (which has $u_i \leq \gamma\rho u_{i-1} + (1 + \gamma)\epsilon_{\text{sub}}$ too, cf. Lemma B.9) requires the same condition. The *advantage* of ICBQ in this regime is the same bound-level advantage as in Theorem B.5: the propagated part carries γ^S , while the additive residual remains uniformly bounded as long as $\tau < 1$. Stated explicitly:

Proposition B.21 (Stability gain). *Under Assumptions B.1, B.2, and B.4 with the locally expansive allowance $1 < \rho < 1/\gamma$, both the sequential and interleaved upper bounds are finite uniformly in depth. Moreover, if $C(L, K, \rho, \gamma)\epsilon_{\text{sub}} \leq \delta \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}}$ for some $\delta \geq 0$ and $\mathcal{B}_L^{\text{seq}} > 0$, then the ratio of the sequential upper bound to the ICBQ upper bound is at least $\gamma^{-S(L,K)}/(1 + \delta)$.*

Proof. Since $\gamma \in (0, 1)$, the condition $1 < \rho < 1/\gamma$ is equivalent to $\tau = \gamma\rho < 1$ with locally expansive teacher blocks. Therefore $(1 - \tau^{L-1})/(1 - \tau) \leq 1/(1 - \tau)$ uniformly in L , and

$$\mathcal{B}_L^{\text{seq}} = \rho(1 + \gamma) \frac{1 - \tau^{L-1}}{1 - \tau} \epsilon_{\text{sub}} + \epsilon_{\text{sub}} \leq \left(\frac{\rho(1 + \gamma)}{1 - \tau} + 1 \right) \epsilon_{\text{sub}}, \quad (70)$$

which is independent of L . Theorem B.5 together with Eq. (26) gives

$$\mathcal{B}_L^{\text{int}} := \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma) \epsilon_{\text{sub}} \leq \mathcal{B}_L^{\text{seq}} + \left(1 + \frac{(1 + \gamma + \gamma^2)\rho}{1 - \tau} \right) \epsilon_{\text{sub}}, \quad (71)$$

where we used $\gamma^{S(L,K)} \leq 1$ and the depth-uniform bound in Eq. (26) on C . Hence the interleaved upper bound is also finite uniformly in L . For the ratio claim, suppose the displayed residual condition holds. Then from Eq. (24) and the definition of $\mathcal{B}_L^{\text{int}}$ in Corollary B.8,

$$\mathcal{B}_L^{\text{int}} = \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} + C(L, K, \rho, \gamma) \epsilon_{\text{sub}} \leq (1 + \delta) \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}}. \quad (72)$$

Since $C(L, K, \rho, \gamma)\epsilon_{\text{sub}} \geq 0$, $\mathcal{B}_L^{\text{seq}} > 0$, and $\gamma^{S(L,K)} > 0$, we also have $\mathcal{B}_L^{\text{int}} \geq \gamma^{S(L,K)} \mathcal{B}_L^{\text{seq}} > 0$, so both sides of the displayed inequality are strictly positive. Dividing by the positive quantity

$\mathcal{B}_L^{\text{int}} \cdot (1 + \delta)\gamma^{S(L,K)}$ and rearranging,

$$\frac{\mathcal{B}_L^{\text{seq}}}{\mathcal{B}_L^{\text{int}}} \geq \frac{1}{(1 + \delta)\gamma^{S(L,K)}} = \frac{\gamma^{-S(L,K)}}{1 + \delta}, \quad (73)$$

which is the claimed ratio bound. $\square$

The main experiments exhibit a related qualitative pattern: Sequential CBQ produces PPL $> 10^4$ for Qwen3-8B in one reported setting, whereas ICBQ yields PPL below 50 for both Qwen3-8B and Llama-3-8B. These observations are consistent with the schedule mechanism but do not estimate the theorem’s uniform constants or validate the bound quantitatively.

B.11 Empirical measurement of γ

We diagnose the per-pair contraction empirically on Llama-2-7B by recording, after each of the 32–1 = 31 two-block refinement calls with ternary DBF, the square-root post-over-pre MSE ratio. The corresponding histogram has mean $\bar{\gamma} \approx 0.67$ and maximum $\bar{\gamma}_{\max} \leq 0.92$.

Uniform vs. aggregate contraction. The quantity $\bar{\gamma}$ reported here is an aggregate diagnostic: it is the average square-root post-over-pre MSE ratio over the CBQ windows actually executed on the calibration set, hence it summarizes how the implemented optimizer behaves *in aggregate*. It is *not*, by itself, the uniform per-sample contraction constant required by Assumption B.2. A theorem-level certificate would instead require a uniform estimate such as

$$\hat{\gamma}_{\text{unif}}(\epsilon_{\text{opt}}) = \max_{\substack{j,n: \\ r_{j,\text{in}}^{(n)} > 0}} \frac{(r_{j,\text{out}}^{(n)} - \epsilon_{\text{opt}})_+}{r_{j,\text{in}}^{(n)}}, \quad (74)$$

where j ranges over the actual CBQ calls of the schedule and n over the calibration samples in each call’s batch, and $r_{j,\text{in}}^{(n)}, r_{j,\text{out}}^{(n)}$ are the pre- and post-pass midpoint residuals of call j on sample n . With this convention, $\hat{\gamma}_{\text{unif}}(\epsilon_{\text{opt}})$ is the smallest γ that makes Assumption B.2 hold uniformly on the actually visited finite sets, given a chosen fixed-point radius ϵ_{opt} .

Reading of Theorem B.5. Using $\bar{\gamma}$ or $\bar{\gamma}_{\max}$ in Corollary B.8 therefore produces only a *schedule-level diagnostic* of the seam mechanism, not a formal certificate. With $L = 32$ and $K = 4$ (so $S = 7$), the aggregate-level diagnostics give an amortization-gain proxy of $\bar{\gamma}^{-7} \approx 16\times$ at the mean and $\bar{\gamma}_{\max}^{-7} \approx 1.8\times$ at the aggregate maximum. The empirical per-model range 1.08–14.5 $\times$ reported in the main experiments sits between these proxies, consistent with the formal γ in Theorem B.5 being some uniform value between $\bar{\gamma}$ and $\bar{\gamma}_{\max}$. Establishing $\hat{\gamma}_{\text{unif}}(\epsilon_{\text{opt}})$ at scale would require logging per-sample midpoint residuals at every CBQ call and is left to future work; the present empirical evidence supports the qualitative validity of the schedule mechanism rather than a quantitative theorem-level certificate.

B.12 Numerical simulation

A standalone numpy script reproducing the rate of Theorem B.5 on a 64-block scalar recurrence toy is included in the supplementary material; running it prints a table of $E_c^{\text{int}}/E_c^{\text{seq}}$ vs. c that matches γ^c to four significant digits over $c \in \{1, \dots, 15\}$.

C Experimental and Implementation Details

C.1 Experimental setup details

Base models and references. The seven base models used in the main experiments are Llama-3.2-3B [Meta Llama Team, 2024], Llama-2-7B [Touvron et al., 2023], Mistral-7B [Jiang et al., 2023], Qwen3-8B [Yang et al., 2025], Llama-3-8B [Grattafiori et al., 2024], Llama-2-13B [Touvron et al., 2023], and Qwen3-14B [Yang et al., 2025].

Quantization and optimization settings. The ternary DBF runs use progressive prefit (50 steps, float32, AdamW). For models below 7B, we use `nsamples = 128` and `dbf.itors = 200`; for models at 7B and above, we use `nsamples = 256` and `dbf.itors = 400`. GPTQ runs use W3 and W2g128 with prefit disabled, GPTQ block size 128, and Hessian damping 0.01. The integer-bit comparison uses the same weight-only regime as the ternary DBF table, with the inner quantizer replaced by GPTQ. No adapter/LoRA parameters are introduced.

Evaluation protocol and benchmarks. Perplexity is measured with the `lm-evaluation-harness` protocol [Gao et al., 2021]. Zero-shot evaluation uses BoolQ, PIQA, HellaSwag, WinoGrande, ARC-e, ARC-c, and OpenBookQA [Clark et al., 2019, Bisk et al., 2020, Zellers et al., 2019, Sakaguchi et al., 2021, Clark et al., 2018, Mihaylov et al., 2018]. Main-text tables focus on C4-calibrated reporting, while supplementary tables include additional calibration views where relevant.

C.2 Calibration bias

WikiText-2 calibration on base models with non-standard attention, such as Qwen3 with grouped-query attention [Yang et al., 2025] or Gemma-3 with local sliding-window attention [Gemma Team et al., 2025], can cover a narrow distribution of positional indices. Under ternary DBF, this restricted calibration distribution can produce a mismatch that the local CBQ loss does not fully capture. In our setting, C4 calibration at the same sample count gave more stable results on the deeper models, making it the default for the main ternary DBF results.

C.3 Implementation and hyperparameters

Hardware. All runs on $1 \times$ NVIDIA H100-80GB. Memory peak with ternary DBF: 32 GB for 7B, 55 GB for 13B models.

Peak memory: ICBQ vs. CBQ. The same practical memory regime is observed under Sequential CBQ and ICBQ: in our implementation, the extra state introduced by ICBQ over a single-sweep CBQ is the passive re-roll buffer `original_inps`, whose size is set by calibration workload and is independent of depth-related CBQ scheduling variables (e.g., L and windowing order for fixed workload). Instantiating the stored-input buffer size at our operating points gives:

Model	d_{hidden}	N	T	dtype	$\Delta\text{mem}_{\text{ICBQ}}$
TinyLlama-1.1B	2048	128	2048	fp16	≈ 1.0 GB
Llama-2-7B / Llama-3-8B / Qwen3-8B / Mistral-7B	4096	256	2048	fp16	≈ 4.3 GB
Llama-2-13B / Qwen3-14B	5120	256	2048	fp16	≈ 5.2 GB

The remaining models use smaller hidden widths ($d_{\text{hidden}} \in \{1024, 1152, 2304\}$), which gives proportionally smaller ICBQ overheads (approximately 0.5, 0.6, and 1.2 GB at $N = 128$, $T = 2048$, fp16). In these cases, the overhead is below 15% of the ternary DBF peak quoted above. Because ICBQ never materializes two active two-block windows simultaneously, the practical peak does not enter a new regime when switching from Sequential CBQ to ICBQ. Measured run logs show similar peak GPU memory across paired settings, consistent with the L, K -independent peak-memory bound up to implementation-level allocator effects.

Wall-clock time: ICBQ vs. CBQ (C4 calibration). To complement the peak-memory comparison, Table 5 reports end-to-end wall-clock runtime for the C4-calibrated ternary DBF runs, comparing Sequential CBQ to ICBQ on eleven models. The mean wall-clock factor is $1.21 \times$. This quantifies the added refinement computation associated with the quality differences reported in the main experiments.

Optimization (ternary DBF runs). Prefit: AdamW, $\eta_{\text{pre}} = 10^{-4}$, 50 steps, float32. DBF: internal iterations $\{200, 400\}$ for $< 7\text{B} / \geq 7\text{B}$. CBQ refinement: Adam with the repository’s optimization defaults (`cbq_epochs = 20`, `cbq_lr =  $5 \times 10^{-5}$` , rollback enabled). Chunk size $K = 4$, with one interleaved refinement sweep per chunk.

Table 5: **Wall-clock comparison on C4-calibrated ternary DBF runs (11 models).** Sequential CBQ versus ICBQ (ours), using Slurm elapsed time. Ratio > 1 indicates ICBQ takes longer.

Model	Seq. CBQ elapsed	ICBQ elapsed	ICBQ/Seq.	Delta
TinyLlama-1.1B	01:20:53	01:27:51	1.09×	+7.0 min
Qwen3-0.6B	01:13:18	01:27:42	1.20×	+14.4 min
Gemma-2-2B	02:11:44	02:43:17	1.24×	+31.6 min
Gemma-3-1B	01:05:13	01:27:08	1.34×	+21.9 min
Llama-3.2-3B	02:06:30	02:32:36	1.21×	+26.1 min
Mistral-7B	08:24:14	09:39:28	1.15×	+75.2 min
Llama-2-7B	09:01:33	10:41:15	1.18×	+99.7 min
Llama-3-8B	08:22:38	09:55:29	1.18×	+92.8 min
Qwen3-8B	09:40:55	11:32:26	1.19×	+111.5 min
Qwen3-14B	15:00:02	19:38:56	1.31×	+278.9 min
Llama-2-13B	15:29:09	18:10:54	1.17×	+161.8 min
Mean (11 models)	—	—	1.21×	—

Optimization (GPTQ runs). Inner quantizer: weight-only GPTQ at either $W3$ (three bits per channel) or $W2g128$ (two bits, group size 128). GPTQ defaults used throughout: block size 128, Hessian damping $\text{percdamp} = 0.01$, no activation reordering ($\text{actorder} = \text{false}$; we observed that enabling actorder changes PPL by < 0.1 and does not interact with the schedule, hence we report with the default value). Prefit is disabled ($T_{\text{pre}} = 0$) for every GPTQ comparison row. The CBQ refinement stage uses the same settings as the ternary DBF runs.

No-prefit ablation setting. The “No-prefit” ablation rows set $T_{\text{pre}} = 0$ while keeping all other hyperparameters identical to those of the DBF Prefit-enabled row (DBF iters, CBQ epochs, learning rate, rollback, chunk size, calibration samples). This is the minimal perturbation that isolates the contribution of the prefit step within the progressive driver.

Calibration. 128 samples of length 2048 from WikiText-2 (or 256 samples for $\geq 7\text{B}$), random starting positions as produced by the shared `get_loaders()` utility. Repository seed default 0 (matching `conf/config.yaml`). Identical calibration splits are shared between DBF and GPTQ runs on the same model.

Reproducibility. Every row of the main ternary DBF results is reproduced with the same calibration pipeline and optimization settings reported above, using chunk size 4, plain CBQ refinement, and 50 prefit steps for ternary DBF runs. The GPTQ rows are produced under the same pipeline with the inner quantizer switched to GPTQ and the bit/group-size pair $(w_{\text{bit}}, w_{\text{group}}) \in \{(3, -1), (2, 128)\}$.

D Supplementary Results

Formatting note. In all tables of this section, **bold** marks the strongest value in each comparison group.

Main-set ternary DBF results under WikiText-2 calibration. Table 6 reports the seven-model DBF counterpart to the main C4-calibrated table, using WikiText-2 calibration only and PPL metrics only (no zero-shot). The qualitative trend is the same: ICBQ is the strongest across all cases for runs calibrated on WikiText-2 as well.

Supplementary DBF PPL results. Table 6 reports supplementary ternary DBF PPL results on four small models in the same format as the main ternary DBF table: WikiText-2 and C4 evaluation under both WikiText-2 and C4 calibration, comparing No-Ref., Sequential CBQ, and ICBQ. As shown in Table 6, the qualitative trend is unchanged: ICBQ is the strongest refinement across these cases.

Ternary-DBF zero-shot results. Table 6 consolidates ternary DBF C4-calibrated zero-shot scores on 11 models with the same three schedules as the main ternary DBF comparison: No-Ref. (ternary DBF only), Sequential CBQ, and ICBQ. As summarized in Table 6, ICBQ most often attains the best score across models and tasks, with a few task-specific ties or Seq. wins.

GPTQ zero-shot results. Table 7 mirrors the ternary DBF zero-shot presentation for GPTQ at $W3$ and $W2g128$ on the main seven-model set. As shown in Table 7, the pattern is consistent with the PPL results: ICBQ is usually best or tied, with a small number of task-specific Seq. wins.

Table 6: Supplementary ternary DBF results. Part a. Main-set ternary DBF results under WikiText-2 calibration (PPL only). Part b. PPL results on 4 small models in the same format as the main ternary DBF results (WikiText-2 and C4, two calibration settings, and three schedules). Part c. Zero-shot results on 11 models under C4 calibration. Zero-shot values are normalized accuracy when available, otherwise accuracy (higher is better).

Part a. Main-set ternary DBF results under WikiText-2 calibration (PPL only).												
Metric	No-Ref. (ternary DBF only)				Seq. CBQ				ICBQ (ours)			
	PPL (Wiki-2)		PPL (C4)		PPL (Wiki-2)		PPL (C4)		PPL (Wiki-2)		PPL (C4)	
Standard models												
Mistral-7B	44.27		329.91		16.24		53.91		10.09		25.59	
Llama-2-7B	11.23		20.05		8.38		14.59		7.78		13.30	
Llama-2-13B	7.63		12.69		6.55		11.15		6.32		10.59	
Deep and non-standard-attention models												
Llama-3.2-3B	40.24		66.50		25.30		49.90		22.98		46.04	
Llama-3-8B	46.12		87.19		27.16		61.21		24.01		48.92	
Qwen3-8B	4661.50		diverge		281.45		8275.66		18.95		37.68	
Qwen3-14B	512.82		2522.21		35.24		119.40		15.52		34.68	
Part b. Supplementary DBF PPL results on 4 small models.												
Calibration data	No-Ref. (ternary DBF only)				Seq. CBQ				ICBQ (ours)			
	Wiki-2		C4		Wiki-2		C4		Wiki-2		C4	
PPL dataset	Wiki-2	C4	Wiki-2	C4	Wiki-2	C4	Wiki-2	C4	Wiki-2	C4	Wiki-2	C4
Qwen3-0.6B	89.84	215.91	165.73	143.15	62.58	137.71	101.4	106.7	54.64	109.88	88.0	92.2
Gemma-3-1B	198.94	407.36	462.94	278.29	86.08	175.41	171.9	130.7	74.02	144.19	140.3	111.9
TinyLlama-1.1B	27.97	47.90	58.03	43.81	18.84	33.58	30.2	30.4	17.04	30.51	25.8	26.7
Gemma-2-2B	85.36	151.92	142.18	92.71	85.31	164.43	106.4	89.2	79.62	149.23	99.1	86.1
Part c. Ternary-DBF zero-shot results on 11 models (C4 calibration).												
Model	Schedule	BoolQ	PIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA				
Qwen3-0.6B	No-Ref.	0.4180	0.5702	0.2925	0.5075	0.3245	0.2159	0.2500				
	Seq. CBQ	0.5728	0.5642	0.2995	0.5178	0.3194	0.2048	0.2520				
	ICBQ (ours)	0.5942	0.5637	0.3071	0.5178	0.3304	0.2167	0.2660				
Gemma-3-1B	No-Ref.	0.5061	0.5533	0.2775	0.5114	0.3220	0.2201	0.2660				
	Seq. CBQ	0.4826	0.5740	0.2813	0.5028	0.3283	0.2065	0.2680				
	ICBQ (ours)	0.4979	0.5811	0.2928	0.5067	0.3375	0.2065	0.3100				
TinyLlama-1.1B	No-Ref.	0.5862	0.6110	0.3368	0.5272	0.3497	0.2227	0.2860				
	Seq. CBQ	0.5832	0.6295	0.3601	0.5249	0.3801	0.2406	0.3060				
	ICBQ (ours)	0.5905	0.6376	0.3725	0.5257	0.3838	0.2551	0.2800				
Gemma-2-2B	No-Ref.	0.5832	0.6502	0.3568	0.5501	0.4419	0.2474	0.2740				
	Seq. CBQ	0.5890	0.6420	0.3529	0.5367	0.4432	0.2440	0.2740				
	ICBQ (ours)	0.6266	0.6485	0.3779	0.5391	0.4739	0.2517	0.3040				
Llama-3.2-3B	No-Ref.	0.6517	0.6202	0.3937	0.5478	0.3855	0.2543	0.2840				
	Seq. CBQ	0.6673	0.6534	0.4253	0.5462	0.4314	0.2765	0.2940				
	ICBQ (ours)	0.6774	0.6801	0.4392	0.5604	0.4815	0.2730	0.2880				
Mistral-7B	No-Ref.	0.3856	0.5887	0.2913	0.5264	0.3022	0.2227	0.2620				
	Seq. CBQ	0.5131	0.6697	0.4524	0.5446	0.4508	0.2858	0.2940				
	ICBQ (ours)	0.6875	0.7236	0.5855	0.5825	0.5568	0.3328	0.3140				
Llama-2-7B	No-Ref.	0.5807	0.6817	0.5259	0.5264	0.4928	0.2679	0.3240				
	Seq. CBQ	0.6407	0.7138	0.5708	0.5785	0.5564	0.3131	0.3580				
	ICBQ (ours)	0.6914	0.7203	0.5875	0.6014	0.5880	0.3345	0.3500				
Llama-3-8B	No-Ref.	0.6734	0.6572	0.4305	0.5580	0.4314	0.2500	0.3000				
	Seq. CBQ	0.7064	0.6931	0.4810	0.5485	0.4966	0.2944	0.2980				
	ICBQ (ours)	0.6963	0.7002	0.5038	0.5770	0.5299	0.3251	0.3180				
Qwen3-8B	No-Ref.	0.3862	0.5180	0.2606	0.4901	0.2567	0.2688	0.2660				
	Seq. CBQ	0.5606	0.5876	0.3184	0.4901	0.3291	0.2526	0.2840				
	ICBQ (ours)	0.7407	0.6910	0.5057	0.5904	0.6111	0.3763	0.3440				
Llama-2-13B	No-Ref.	0.7505	0.7307	0.6261	0.6322	0.6662	0.3746	0.3920				
	Seq. CBQ	0.8083	0.7938	0.6012	0.7206	0.7950	0.4829	0.3500				
	ICBQ (ours)	0.8080	0.7927	0.6016	0.7151	0.7955	0.4855	0.3520				
Qwen3-14B	No-Ref.	0.3786	0.5408	0.2632	0.4964	0.2976	0.2278	0.2660				
	Seq. CBQ	0.8933	0.7987	0.6103	0.7301	0.8413	0.5862	0.3500				
	ICBQ (ours)	0.8930	0.8020	0.6087	0.7301	0.8426	0.5896	0.3440				

Table 7: **GPTQ zero-shot results on the main seven-model set (C4 calibration).** Weight-only GPTQ, no prefit ($T_{\text{pre}} = 0$), and the same schedules as the main GPTQ comparison: GPTQ only, Sequential CBQ, and ICBQ. Values are normalized accuracy when available, otherwise accuracy (higher is better).

Model	Bit-width	Schedule	BoolQ	PIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA
Mistral-7B	W3	GPTQ only	0.7615	0.7965	0.7580	0.6882	0.7088	0.4369	0.3980
		Seq. CBQ	0.7865	0.8074	0.7738	0.6803	0.7588	0.4855	0.4280
		ICBQ (ours)	0.7817	0.8009	0.7703	0.6756	0.7319	0.4667	0.4120
	W2g128	GPTQ only	0.6349	0.7258	0.5905	0.5580	0.5501	0.3114	0.3140
		Seq. CBQ	0.7269	0.7889	0.7134	0.6385	0.6776	0.3857	0.3880
		ICBQ (ours)	0.7131	0.7851	0.7137	0.6322	0.6582	0.3908	0.3960
Llama-2-7B	W3	GPTQ only	0.7318	0.7671	0.6937	0.6377	0.6414	0.3848	0.3800
		Seq. CBQ	0.7394	0.7769	0.7176	0.6535	0.6974	0.4061	0.4420
		ICBQ (ours)	0.7483	0.7797	0.7242	0.6472	0.7125	0.4241	0.4240
	W2g128	GPTQ only	0.5367	0.6937	0.5631	0.5643	0.4912	0.2824	0.3420
		Seq. CBQ	0.6618	0.7508	0.6435	0.6117	0.6128	0.3396	0.3920
		ICBQ (ours)	0.6691	0.7601	0.6565	0.6148	0.6178	0.3575	0.3980
Llama-2-13B	W3	GPTQ only	0.7572	0.7807	0.7482	0.6930	0.7071	0.4369	0.4300
		Seq. CBQ	0.7627	0.7894	0.7551	0.7024	0.7205	0.4514	0.4420
		ICBQ (ours)	0.7517	0.7943	0.7509	0.6977	0.7260	0.4522	0.4540
	W2g128	GPTQ only	0.6645	0.7301	0.6378	0.6188	0.5981	0.3311	0.3520
		Seq. CBQ	0.7474	0.7693	0.6856	0.6535	0.6511	0.4002	0.4100
		ICBQ (ours)	0.7517	0.7720	0.6913	0.6638	0.6540	0.3899	0.4020
Llama-3.2-3B	W3	GPTQ only	0.6587	0.6415	0.5842	0.6054	0.4280	0.2884	0.3200
		Seq. CBQ	0.7505	0.7225	0.6434	0.5975	0.6023	0.3703	0.3500
		ICBQ (ours)	0.6570	0.7356	0.6570	0.6022	0.6439	0.4053	0.3480
	W2g128	GPTQ only	0.4251	0.5528	0.3584	0.5036	0.3190	0.2270	0.2520
		Seq. CBQ	0.6456	0.6730	0.5141	0.5501	0.4802	0.2986	0.3180
		ICBQ (ours)	0.6627	0.6725	0.5420	0.5777	0.5021	0.3038	0.3260
Llama-3-8B	W3	GPTQ only	0.6752	0.6289	0.6303	0.6212	0.4091	0.2577	0.3380
		Seq. CBQ	0.7771	0.7579	0.6442	0.6748	0.7134	0.4497	0.3900
		ICBQ (ours)	0.7829	0.7622	0.6925	0.6890	0.7218	0.4633	0.3980
	W2g128	GPTQ only	0.4700	0.5631	0.3780	0.5138	0.3375	0.2184	0.2400
		Seq. CBQ	0.6700	0.7165	0.5959	0.6006	0.5720	0.3225	0.3460
		ICBQ (ours)	0.6939	0.7399	0.6241	0.6062	0.6153	0.3592	0.3640
Qwen3-8B	W3	GPTQ only	0.7303	0.7416	0.6509	0.5833	0.6237	0.3737	0.3780
		Seq. CBQ	0.8086	0.7628	0.6937	0.6504	0.7521	0.5000	0.4080
		ICBQ (ours)	0.8168	0.7661	0.7092	0.6504	0.7765	0.5290	0.4040
	W2g128	GPTQ only	0.5196	0.6431	0.4550	0.5114	0.4335	0.2688	0.2720
		Seq. CBQ	0.7639	0.7454	0.6394	0.6377	0.7214	0.4659	0.3860
		ICBQ (ours)	0.7722	0.7514	0.6567	0.6393	0.7277	0.4548	0.4100
Qwen3-14B	W3	GPTQ only	0.8382	0.7682	0.7291	0.6764	0.7597	0.4872	0.4440
		Seq. CBQ	0.8624	0.7905	0.7601	0.7072	0.7980	0.5478	0.4680
		ICBQ (ours)	0.8630	0.7922	0.7621	0.7080	0.7992	0.5563	0.4520
	W2g128	GPTQ only	0.7122	0.7209	0.6027	0.5620	0.5892	0.3575	0.3240
		Seq. CBQ	0.8245	0.7731	0.7080	0.6969	0.7858	0.5367	0.4260
		ICBQ (ours)	0.8162	0.7764	0.7158	0.6969	0.7753	0.5119	0.4400